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Stock and Recruitment[1]

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ABSTRACT

Plotting net reproduction (reproductive potential of the *adults* obtained) against the density of stock which produced them, for a number of fish and invertebrate populations, gives a domed curve whose apex lies above the line representing replacement reproduction. At stock densities beyond the apex, reproduction declines either gradually or abruptly. This decline gives a population a tendency to oscillate in numbers; however, the oscillations are damped, not permanent, unless reproduction decreases quite rapidly *and* there is not too much mixing of generations in the breeding population. Removal of part of the adult stock reduces the amplitude of oscillations that may be in progress and, up to a point, *increases* reproduction.

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INTRODUCTION

GENERAL

There exists today a considerable body of knowledge which goes by the name of "the theory of fishing" or "the modern theory of fishing"—the work of a succession of the most distinguished fishery biologists of our time. It is concerned mainly with predicting what catch can be obtained from a given number of young fish recruited to a fishery, if their initial size and the growth and natural mortality rates prevailing are known. That is, methods have been developed for computing the effects of different rates of exploitation, of changes in rate of exploitation from year to year, of different minimum size limits, etc., upon the yield obtained. Not only that, but much progress has been made in developing methods of determining the actual magnitudes of the population statistics required to make these calculations.

Valuable as the above contributions have been, they comprise only half of the biological information needed to assess the effects of fishing and an optimum level of exploitation. Fishing changes the absolute and the relative abundance of mature fish in a stock, and the effect of this upon the number of recruits in future years has often been considered only in the most general manner. The points of view encountered usually range from an assumption of direct proportion between size of adult stock and number of recruits, to the proposition that number of recruits is, for practical purposes, independent of the size of the adult stock. The possibility of a *decrease* in recruitment at higher stock densities has less often been considered.

The scarcity of information on this subject is quite explicable, since it usually requires many years of continuous observation to establish a relation between size of stock and the number of recruits which it produces. However, it has become an urgent problem to have a scientific description of the regulation of abundance of fish stocks, in order to complete the basis for predicting optimum levels of exploitation. This paper attempts to summarize some of the theoretical and factual information available, both from fish populations and from other animals, and to provide a stimulus to studies which will eventually put the subject on a solid foundation.

THEORY OF POPULATION REGULATION

Basic in any stock-recruitment relationship is the fact that a fish population, even when not fished, is limited in size; that is, it is held at some more or less fluctuating level by natural controls. Ideas concerning the nature of such controls were first clarified and systematized by the Australian entomologist Nicholson

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(1933). He showed that, while the level of abundance attained by an animal can be affected by any element of the physical or biological environment, the immediate mechanism of control must always involve competition, using that word in a broad sense to include any factor of mortality whose effectiveness increases with stock density[2]. The term *density-dependent* mortality was used for the same concept by Smith (1935). More strictly, density-dependent causes of mortality should include both those which become more effective as density increases and those which become less so. The former are the ones which provide control of population size; and they have been called *concurrent* (Solomon, 1949), *compensatory* (Neave, 1953) or *negative* (Haldane, 1953). The opposed terms are *inverse, depensatory* and *positive,* all referring to density-dependent factors which become *less* effective as density increases.

No sharp line can be drawn between the kinds of mortality which are compensatory and those which are not, although Nicholson, Smith and others have felt that as a rule biological factors tend to predominate among the former, and physical agents among the latter, for insects at least. Among fishes, extremes of water temperature, drought and floods, are physical agents which may often cause mortality whose effectiveness is independent of stock density; whereas deaths from such biological causes as disease, parasitism, malnutrition and predation will usually become relatively more frequent as stock density increases. Yet exceptions to the rule above are sufficiently numerous to make the rule itself of doubtful applicability to fishes. For example, the biological factor of predation may have a uniform effectiveness over a considerable range of prey abundance, or at times may even become more effective at lower prey densities; and most if not all physical causes of mortality are compensatory when stock becomes dense enough that some of its members are forced to live in exposed or unsuitable environments. In addition, it is of course often difficult to ascribe a death to any single cause.

There is no necessary relation between the relative magnitudes of the causes of mortality existing at a given time, as measured by the fraction of the stock which each kills, and their relative contribution to compensation. An important and deadly agent of mortality may be strongly density-dependent, or weakly so, or not at all; and different agents may have their maximum compensatory effect over quite different ranges of density.

Age Incidence of Compensatory Mortality

Density-dependent causes of mortality could affect the abundance of either the existing adult stock or the young which it produces. In this paper we will

[2]This almost axiomatic proposition is implied in the writing of various earlier authors back as far as Malthus, but Nicholson was the first to formulate it explicitly and to emphasize its importance: Haldane (1953) calls his inspiration "a blinding glimpse of the obvious". The theorem and its diverse consequences were elaborated mathematically by Nicholson and Bailey (1935). Subsequent writers have developed it with varying emphasis, but nothing very substantial seems to have been added. Solomon (1949) gives a useful review of this literature, and Varley's (1947) quantitative assessment of various agents controlling the abundance of a trypetid fly population is outstanding.

consider mainly the effects of density dependence in the mortality which strikes the younger members of a population—among fishes, the eggs, larvae, fry and fingerlings. That is, the relative abundance of a brood will be considered to be determined by the time the first of its female members begin to mature: subsequent mortality is assumed to be non-compensatory.

This distinction between immature and mature stages of the life history, and the restriction of compensatory mortality to the former, is the principal difference between the thesis of this paper and that of earlier treatments of effects of density upon reproduction (e.g., Hutchinson, 1948; Haldane, 1953; Fujita and Utida, 1953; and many others). These have usually taken the Verhulst logistic equation as a point of departure:

$$\frac{dN_t}{dt} = \frac{bN_t(K - N_t)}{K}$$

(N_t is abundance at time t; K is equilibrium abundance; b is the instantaneous rate of increase of the population at densities approaching zero.) This expression implies a continuous tendency for the population to adjust itself toward the equilibrium size K, whether it is currently less than or greater than K. The adjustment would evidently have to involve compensatory mortality among the adult stock when its density is greater than K. Though such mortality is not impossible, there is in fish populations, at least, little indication of it; and as a matter of fact this aspect of the logistic equation has never been applied to any concrete biological situation.

Our assumption that *no* compensatory mortality occurs among the mature stock is unlikely to be strictly true of any species, and may not be even approximately true of some. Nevertheless it has seemed worth while to follow out the consequences of making this distinction between mature and immature, with fishes particularly in mind. The opportunities for compensatory effects are so much greater during the small, vulnerable early stages of a fish's life, that restriction of compensation to those stages seems likely to have wide applicability as a useful approximation.

To further simplify the initial approach to the problem, fishing is considered to attack only mature individuals, so that the *recruitment* produced by a given density of mature stock means both the number of commercial-sized fish, and the number of maturing fish, which result from its reproductive activity. The term *reproduction* will be used in a similar but slightly more general sense, to mean the number of young surviving to any specified age after compensation is practically complete; it will *not* mean the initial number of eggs or newborn young.

The above conditions and definitions will be assumed to apply unless exception is made specifically.

Kinds of Population Control Mechanisms

Compensatory types of mortality can be of various kinds. Some of the more likely possibilities are as follows:

1. Prevention of breeding by some members of large populations because

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all breeding sites are occupied. Note that territorial behaviour may restrict the number of sites to a number less than what is physically possible.

2. Limitation of *good* breeding areas, so that with denser populations more eggs and young are exposed to extremes of environmental conditions, or to predators.

3. Competition for living space among larvae or fry, so that some individuals must live in exposed situations. This too is often aggravated by territoriality—that is, the preemption of a certain amount of space by an individual, sometimes more than is needed to supply necessary food.

4. Death from starvation or indirectly from debility due to insufficient food, among the younger stages of large broods, because of severe competition for food.

5. Greater losses from predation among large broods because of slower growth caused by greater competition for food. It can be taken as a general rule that the smaller an animal is, the more vulnerable it is to predators, and hence any slowing up of growth makes for greater predation losses. Since abundant year-classes of fishes have often been found to consist of smaller-than-average individuals (Hile, 1936), this may well be a very common compensatory mechanism among fishes (cf. Ricker and Foerster, 1948; Johnson and Hasler, 1954).

6. Cannibalism: destruction of eggs or young by older individuals of the same species. This can operate in the same manner as predation by other species, but it has the additional feature that when eggs or fry are abundant the adults which produced them tend to be abundant also, so that percentage destruction of the (initially) denser broods of young automatically goes up—provided the predation situation approaches the type in which kills are made at a constant fraction of random encounters (cf. page 609, below).

7. Larger broods may be more affected by macroscopic parasites or microorganisms, because of more frequent opportunity for the parasites to find hosts and complete their life cycle.

8. In limited aquatic environments there may be a "conditioning" of the medium by accumulation of waste materials that have a depressing effect upon reproduction, increasingly as population size increases.

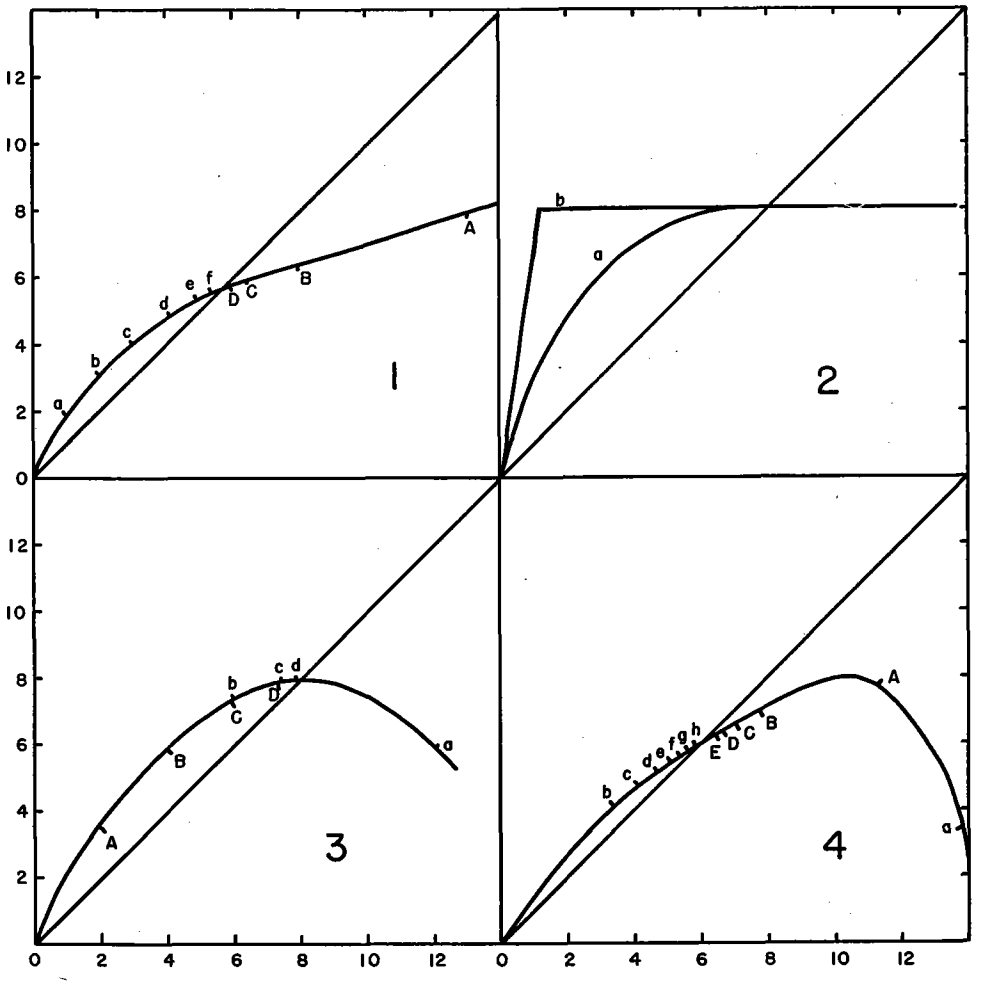
Not all of the above compensatory effects need exist, or be important, in any given population, or in the same population every year. To a considerable extent they are likely to be complementary, so that if, for example, exceptionally favourable conditions permitted a good hatch of even a large spawning of eggs, a reduced growth rate of the fry would permit increased predation and so reduce survival in that way.

TYPES OF REPRODUCTION CURVES

Whatever the various kinds of compensatory and non-compensatory mortality acting on a brood may be, the average resultant of their action, over the existing range of environmental conditions, is represented by the average size of maturing brood, or recruitment, which each stock density produces. A graph of this relationship between an existing stock, and the future stock which the existing

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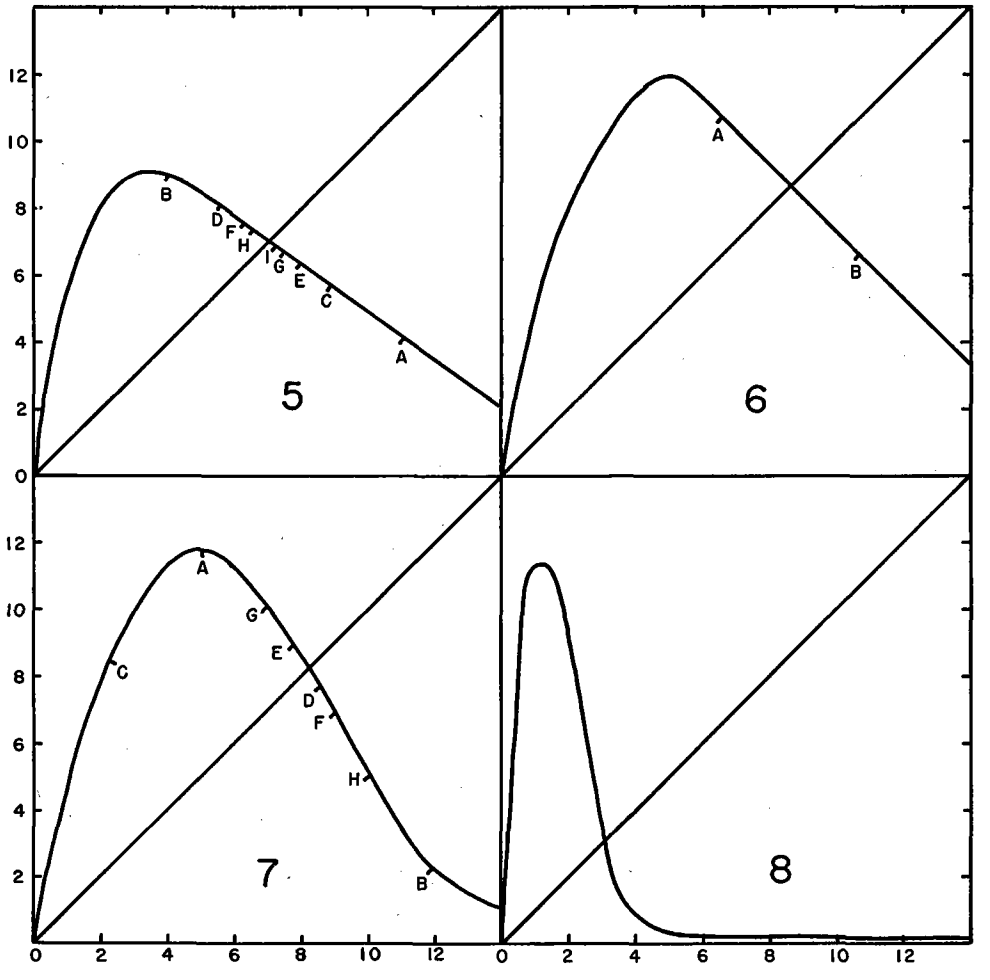


FIGURES 1–4. Stock-reproduction relationships characterized by a stable equilibrium. *Abscissa*— number of eggs produced by parent stock in a given year; *ordinate*—number of eggs produced by the progeny of that year.

stock produces, will be called a "reproduction curve". It is most convenient to label the axes in terms of the *eggs* in present and future generations, respectively[3]. The abscissa represents the mature eggs produced by the current year's stock. The ordinate represents the total of mature eggs produced by the progeny re-

[3]The argument is developed here in terms of populations of oviparous fishes which spawn once a year, but it can readily be modified to apply to other kinds of animals; an example for a viviparous animal is given on page 596. Among mammals, choice of the most suitable census age for plotting on reproduction curves may require care. If newborn young are used, effects of stock density upon frequency of conception and uterine mortality may be overlooked. If number of mature females is used, it should preferably be adjusted to take care of age variation in litter size or frequency, as the age structure of the population changes.

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FIGURES 5–8. Stock-reproduction relationships in which there is an oscillating equilibrium, or (in 5) an oscillating approach to stable equilibrium. Axes as in Figures 1–4.

sulting from the current year's reproduction (obtained by summing over such period of time as the current year's hatch is a component of future years' stocks). In a state of nature, or with a stable fishery, the average size of parental and filial egg production, defined as above, tends to be equal over any long period of years, although striking changes may occur between individual years, or generations.

Figures 1–8 show a number of possible types of reproduction curve. In each of them the straight diagonal constitutes a useful boundary of reference which will be called the "45-degree line". Any curve lying wholly above this line describes a stock which is increasing without limit, hence such a curve cannot exist in practice. Similarly a curve below the 45-degree line describes a stock

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that will decrease to zero in a few generations[4]. The 45-degree line itself would describe a stock in which density dependence is absent, the filial generation tending always to be equal to the parental, except as factors independent of density deflect it. Such a stock would have no mechanism for the regulation of its numbers: if only density-independent causes of mortality exist, the stock can vary without limit, and must eventually by chance decrease to zero. Thus the first qualification of a reproduction curve is that it must cut the 45-degree line at least once—usually only once—and must end below and to the right of it.

Figures 1–8 indicate some of the types of recruitment curve that might exist in actual populations. All are characterized by having a region above the 45-degree line in which reproduction is more than adequate to replace the existing stock, and a region below the 45-degree line in which reproduction is inadequate to replace existing stock.

In the population of Figure 1, rate of reproduction (ratio of filial to parental eggs) decreases continuously as size of stock increases, although the slope of the curve becomes stabilized soon after the 45-degree line is crossed, and the actual number of young produced continues to rise indefinitely. Beyond the 45-degree line, however, this number is inadequate to fully replenish the stock.

In the population of Figure 2a, rate of reproduction also declines continuously; the actual number of recruits produced reaches an asymptotic level and thereafter does not change. Curve 2b differs from 2a in that it rises more steeply and it is initially straight; i.e., at low stock densities rate of reproduction is large, and constant. The range of densities over which there is constant recruitment is quite broad in b, simulating the condition loosely described by the statement "there are always as many recruits as the grounds can support".

Figures 3–8 show reproduction curves in which numbers of recruits begin to decrease after stock reaches some large magnitude. Such curves have an ascending left limb, a dome and a descending right limb. The descending right limb provides of course a more severe control of stock size at the higher densities. The differences in position of the dome and in the slope of the two limbs are of importance in determining changes in stock abundance, as described below.

All these curves are meant to represent the net effect of the sum total of density-dependent mortality factors acting upon the population. The reproduction of any actual year is affected also by density-independent factors, so that the actual number of young produced will deviate from the number indicated by the curve. In a later section the effect of such density-independent factors will be

[4]This statement is likely to be literally true only if the reproduction curve is plotted and fitted on logarithmic axes. On arithmetic axes a curve lying wholly below the 45-degree line *could* describe a stable population (having more than one age in the breeding stock) provided there were moderate to large random deviation from the average relationship. This point is discussed further on page 578, where it is evident that it would be advantageous to fit reproduction curves on logarithmic axes so that deviations above and below the curve would be in better balance. However such plotting has the serious disadvantage that statements of parent-progeny relationships in terms of the slopes of the two limbs of the curve become too complex to be practical. The best way out of this difficulty would be to fit the curve on a logarithmic plot, then transform it to arithmetic axes for interpretation.

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examined. The first task will be to consider how a population behaves, under the strict control of a reproduction curve, when it is given some initial randomly chosen position.

This subject, and others later, will be considered for two situations separately. The first is where the currently hatched brood will constitute the whole of the breeding stock of a subsequent generation; that is, there is no mixing of ages in the spawning stock. This situation is fairly common among insects; for example, many mayflies, stoneflies, caddis flies, etc., have a single brood per year, usually with a one-year life cycle. Among vertebrates this condition is exceptional, but it exists in some fishes. There the length of life is commonly more than one year, so that two or more separate populations or "lines" exist concurrently.

The other situation to be considered is of course that where two or more broods contribute to the spawning stock at any given moment. Among vertebrates at least, this is the usual state of affairs.

REPRODUCTION IN THE ABSENCE OF DENSITY-INDEPENDENT VARIABILITY

Single-Age Spawning Stocks

In all of the relationships shown in Figures 1–8, the stock is in equilibrium at the density at which the reproduction curve cuts the 45-degree line—that is, the stock is then producing enough progeny, and only enough, to replace its current numbers. In Figure 1, an initial deflection of abundance to either side of the equilibrium point is compensated by a gradual, asymptotic return to equilibrium, for example by paths A–D or a–f. In Figure 2a things work the same way to the left of the equilibrium point; to the right of it any deflection, no matter how great, is returned to equilibrium in a single generation. In Figure 2b deviations to the left, as far as 1.5 units of stock, are also returned immediately to the equilibrium level.

In Figure 3 the equilibrium point is at the top of the dome of the reproduction curve. The curve resembles 1 as regards deflections to the left of the equilibrium point; but a displacement to the right is followed by an immediate return across the 45-degree line to the ascending limb, after which it "climbs" this limb to the equilibrium point (a–d). The curve of Figure 4 is similar, but the dome lies to the right of the 45-degree line; A–E and a–h are possible paths to the equilibrium point.

In Figure 5, a stock deflected from equilibrium to any position along the descending limb will oscillate back and forth about the equilibrium point with decreasing amplitude, for example by the route A–I. If the deflection is great enough, the stock may have to climb the left limb before it is swung over to the right limb and begins the oscillating phase.

In Figure 5 the right limb has a downward (negative) slope numerically less than -1. In Figure 6 this slope is exactly -1 after the dome is passed. Here any moderate deflection along the straight part of the right limb results in a swing back across the 45-degree line of exactly the same magnitude, so that the deflection tends to be perpetuated indefinitely; for example, A and B are two such

conjugate points. The intersection of the 45-degree line is a point of "indifferent" equilibrium; it is itself stable, but there is no inherent tendency for the stock to return to that level.

Finally, when the slope of the right limb of a reproduction curve lies between -1 and $-\infty$, equilibrium at the 45-degree line is not merely indifferent, it is unstable. That is, any deflection from equilibrium, no matter how small, initiates a series of oscillations along the right limb whose amplitude increases until the dome of the curve is reached or surpassed. The latter event usually sends the stock back to the right limb and the cycle begins again. No matter where they begin, all such cycles eventually reach the dome of the curve, and a stable oscillation series is established for which the dome is a convenient starting point. The cycle in Figure 7 is A, B, C, D, E, F, G, H, A, etc. (cf. Fig. 11D, below); the number of stages in this cycle depends upon the exact shape of the curve, chiefly upon whether or not one stage lands close to the 45-degree line. Figure 8 represents a more extreme situation, in which substantial reproduction is obtained over only a narrow range of stock densities considerably below the equilibrium level, and the stock would be subject to violent oscillations.

Multiple-Age Spawning Stocks

When a spawning population consists of two or more age-groups, the young produced in a given year contribute to the stock of more than one future year, and the results of a deflection from equilibrium abundance are much modified. A fairly plausible example is where each brood contributes to the spawning stocks of four future years, in the ratio 2:3:3:2, and first spawning occurs 4 years after a brood was produced (Table I). Figure 9 shows the result of an initial deflection of such a stock to an abundance of 12, on some of the reproduction curves of Figures 1–8. The course of events for the most part reflects what was learned in the single-spawning situation. Those based on Figures 1–6 all end up at the stable equilibrium level; this being reached by direct approach for 1, 2 and 3, with one hesitation for 4, and by a series of damped oscillations for 5 and 6. From Figures 7 and 8 series of undamped oscillations are obtained, that is, permanent cycles of abundance.

It may seem surprising that curve 6 too does not generate permanent oscillations, but the mixing of year-classes gradually brings the stock to a steady level, in the absence of any tendency toward divergence.

The permanent oscillations of the type produced by curves 7 and 8 will repay more extended discussion. Figure 10 depicts series of oscillations, based on Figure 7, but with each brood contributing to the spawning stock for only two years. Time of first maturity is successively delayed one, two and more years, so that the average contribution to reproduction is made 1.5, 2.5, 3.5, etc., years after the brood in question existed as mature eggs. In every case stable cyclical fluctuations exist, just as in Figure 9F. Their *period* is always double the mean interval from egg to egg, that is, 3, 5, 7, etc., years.

The *amplitude* of the cycle varies. If the fish spawn first in the year after their appearance, amplitude is very small, but it quickly increases if maturity

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TABLE I. An illustration of how to obtain the definitive distribution of population abundance for given conditions and a given reproduction curve, in this case Figure 7. Four years are assumed to elapse between a brood's existence as fertilized eggs and its first year of contribution to reproduction, while its total egg production is divided among the 4 successive years of its mature existence in the ratio 2 : 3 : 3 : 2. An arbitrary stock density (measured in terms of its egg production) is taken as a starting point, in this case 2.0 units (column 5). The reproduction corresponding to this abscissal value is read off from the ordinate of Figure 7, namely 7.9 egg units (column 6). These eggs are divided among the 4 years in which they are actually laid, beginning 4 years after the year in which the fish that carry them were hatched. For example, the 10.2 units of "recruitment" in line 5 are divided among line 9 (2.0 units), line 10 (3.1 units), line 11 (3.1 units), and line 12 (2.0 units). Breeding potential for each year (column 5) is obtained by horizontal addition of the contributions of the 4 age-groups in columns 1–4. In this example the definitive 11-year period of oscillation is achieved in the first peak-to-peak interval, while the average definitive amplitude, approximately 3.6 to 11.3, is apparent in the second peak-to-peak interval. Adjustment to the conditions imposed by the reproduction curve is not always as rapid as this.

Eggs produced at successive ages						Eggs produced at successive ages					
IV	V	VI	VII	Stock	Recruitment	IV	V	VI	VII	Stock	Recruitment
0.4	0.6	0.6	0.4	2.0	7.9	0.6	1.3	2.0	2.0	5.9	11.4
0.4	0.6	0.6	0.4	2.0	7.9	0.6	0.8	1.3	1.4	4.1	11.5
0.4	0.6	0.6	0.4	2.0	7.9	1.0	0.9	0.8	0.8	3.5	10.7
0.4	0.6	0.6	0.4	2.0	7.9	1.7	1.4	0.9	0.6	4.6	11.8
1.6	0.6	0.6	0.4	3.2	10.2	2.3	2.6	1.4	0.6	6.9	10.3
1.6	2.4	0.6	0.4	5.0	11.9	2.3	3.4	2.6	1.0	9.3	6.4
1.6	2.4	2.4	0.4	6.8	10.4	2.1	3.4	3.4	1.7	10.6	4.2
1.6	2.4	2.4	1.6	8.0	8.7	2.4	3.2	3.4	2.3	11.3	3.1
2.0	2.4	2.4	1.6	8.4	8.0	2.1	3.5	3.2	2.3	11.1	3.5
2.4	3.1	2.4	1.6	9.5	6.1	1.3	3.1	3.5	2.1	10.0	5.3
2.1	3.6	3.1	1.6	10.4	4.6	0.8	1.9	3.1	2.4	8.2	8.3
1.7	3.1	3.6	2.0	10.4	4.6	0.6	1.3	1.9	2.1	5.9	11.4
1.6	2.6	3.1	2.4	9.7	5.8	0.7	0.9	1.3	1.3	4.2	11.6
1.2	2.4	2.6	2.1	8.3	8.2	1.1	1.0	0.9	0.8	3.8	11.1
0.9	1.8	2.4	1.7	6.8	10.4	1.7	1.6	1.0	0.6	4.9	11.9
0.9	1.4	1.8	1.6	5.7	11.5	2.1	2.5	1.6	0.7	6.9	10.3
1.2	1.4	1.2	1.2	5.2	11.9	2.3	3.1	2.5	1.1	9.0	7.0
1.6	1.7	1.4	0.9	5.6	11.7	2.2	3.5	3.1	1.7	10.5	4.4
2.1	2.5	1.7	0.9	7.2	9.9	2.4	3.3	3.5	2.1	11.3	3.1
2.3	3.1	2.5	1.2	9.1	6.8	2.1	3.6	3.3	2.3	11.3	3.1
2.4	3.5	3.1	1.6	10.6	4.2	1.4	3.1	3.6	2.2	10.3	4.8
2.3	3.6	3.5	2.1	11.5	2.8	0.9	2.1	3.1	2.4	8.5	7.8
2.0	3.5	3.6	2.3	11.4	3.0	0.6	1.3	2.1	2.1	6.1	11.2
1.4	3.0	3.5	2.4	10.3	4.8	0.6	0.9	1.3	1.4	4.2	11.6
0.8	2.0	3.0	2.3	8.1	8.5	1.0	0.9	0.9	0.9	3.7	11.0

is delayed. More generally, appreciable amplitude in cycles of this type depends upon a preponderance of the reproduction of a brood occurring after one or more spawning periods have elapsed from the time of its birth.

With the longer intervals of lag between hatching and first spawning the cycles of Figure 10 become less regular; minor peaks appear, and the pattern is duplicated more closely at 2-cycle intervals, a feature which can be detected also in Figure 9F. However, these tendencies are much less apparent when more than two ages occur in the spawning stock, and only the dominant peaks would be detectable with any combination of average age of spawners and average age at first maturity that is apt to occur in nature.

Endless examples of reproduction-curve cycles can be constructed. Any desired combination of period and amplitude can be obtained, in more than one way, by selecting appropriate combinations of reproduction curve and age distribution of breeding. The general characteristics of such cycles can be summarized as follows:

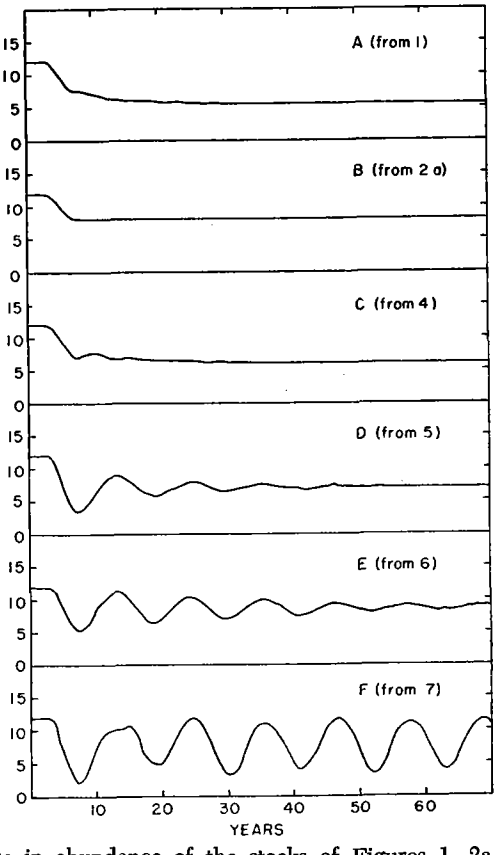


FIGURE 9. Change in abundance of the stocks of Figures 1, 2a and 4–7, following an initial sustained deflection to an abundance of 12 units, when the spawning stock is composed of 4 year-classes and first spawning is in the fourth year after hatching. *Abscissa*—years (generations); *ordinate*—relative abundance (egg production) of the mature stock.

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1. Cycles occur when the outer part of the reproduction curve slopes downward, provided this slope begins at some point above the 45-degree line.

2. Cycles are damped and eventually disappear when the slope of the outer limb of the reproduction curve lies between 0 and —1. They are permanent when the slope is numerically somewhat more than —1, the exact critical limit depending upon the amount of mixing of generations in the spawning stock and the interval to first spawning.

3. Period of oscillation is determined by the mean length of time from parental egg to filial eggs, being twice that interval or close to it. It is independent of the exact shape of the reproduction curve, and also independent of the number of generations in the spawning stock, provided there is more than one.

4. Amplitude of oscillation depends partly on the exact shape of the reproduction curve.

5. Amplitude of oscillation tends to decrease with increase in the number of generations comprising the spawning stock.

6. Amplitude of oscillation increases rapidly with increase in number of generations between parental egg and the first production of filial eggs, up to a

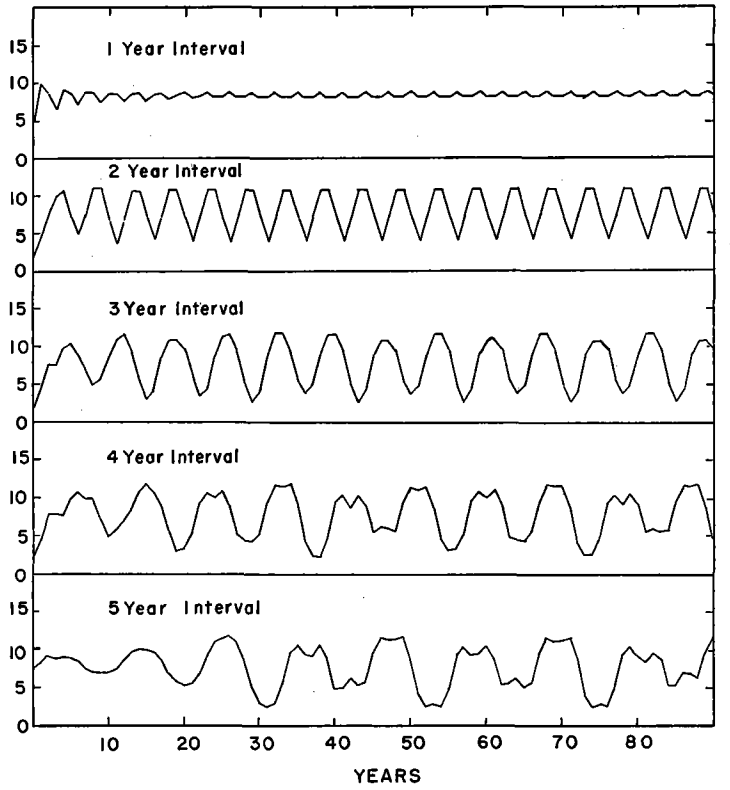


FIGURE 10. Population oscillations determined by the reproduction curve of Figure 7, when there are two ages in the spawning stock and spawning first occurs after 1, 2, 3, 4 and 5 years, respectively, from deposition of the parental eggs. Axes as in Figure 9.

limit imposed by the shape of the reproduction curve. When reproduction by a brood begins strongly in the generation following its birth, the oscillations are so weak that they could not be recognized in practice.

VARIATIONS IN REPRODUCTION PRODUCED BY FACTORS INDEPENDENT OF DENSITY

General

A comparison of density-dependent and density-independent reproduction is desirable in order to find possible means of distinguishing the two by their effects on population abundance, particularly since it has been suggested that some of the apparently periodic variations in animal numbers may reflect random variability alone (Hutchinson, 1948; Palmgren, 1949; Cole, 1951, 1954).

As an introduction, it is known that quantitative events selected completely at random tend to have an average peak-to-peak interval of exactly 3, provided they are classified finely enough that like values do not occur in adjacent positions. Cole (1951) demonstrates this mathematically and found approximately this period in a selection from Tippett's table of random numbers.

Cole uses a sequence of Tippett's numbers as a model with which to compare cycles of animal abundance in nature. Such a model, however, is not appropriate for our present purpose, because it must be interpreted as a reflection of random variation in the capacity of the environment to sustain the animal in question, rather than random variation in number of mature progeny produced per female. This is true for the following reason. When using Tippett's table or any similar assemblage, the number of numbers available is finite. The smallest number can as easily be followed by the largest as by any other; the largest cannot be followed by anything larger. The biological characteristics of a corresponding population are that it must have a very large potential rate of increase (since the greatest possible abundance can follow directly on the least), and it must have severe compensation at the higher stock densities (since they are close to a ceiling of abundance which cannot be exceeded). Few if any populations could meet both these conditions in the course of a single reproductive period, so that Cole's model, to have verisimilitude, must be applied only to situations where a census is taken at intervals of several generations. Thus it is not appropriate, for example, to fish populations when censused yearly, since these usually spawn only once a year.

In any event, a simple series of random numbers is not a suitable model of random variation *in success of reproduction*. What then *is* a suitable model? Since extremes of environmental conditions are less common than conditions approaching normal, our first postulate will be that a normal frequency distribution provides a fairly realistic picture of the relative frequencies of the resultants of the various independent factors making for success or failure in reproduction; from which resultants a random selection is to be made. Table 8.6 of Snedecor (1946) was used for this purpose, selection of t-values being made corresponding to the figure in the ".00" column closest to each of a sequence of selections from Tippett's random table. The series obtained begins as follows: 1.8, 0.5, 0.2, 0.1, 0.2, 0.0,

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1.0, 0.1, etc. Tippett's table was again used to divide these into items representing reproduction above and below the replacement level, respectively; this gave the series below, decreases being followed by *d*.

1.8*d*, 0.5, 0.2, 0.1, 0.2*d*, 0.0, 1.0*d*, 0.1, etc.

The next question is to decide how these are to be applied to an existing reproductive potential. An illustration will be of assistance. Suppose that a breeding population matures 10 billion (10^9) eggs; and let us define "average" environmental conditions here as those which will permit the maintenance level of reproduction, that is, 10 billion future mature eggs produced by the progeny of the current year's sexual activity. Variations in environmental conditions add to or delete from this average production. The most severe conditions imaginable can reduce reproduction to zero. The most favourable conditions possible will permit the survival of all the eggs and thus, assuming equality of the sexes, produce future reproductive units to the number of 5 billion times the total expectation of egg production of a female fish just maturing—which expectation might easily be 2,000,000 in the case of the cod, for example. The mean and the extreme limits of reproduction would thus be:

	Mature eggs	*No. of fish (at first maturity)*
Lower limit:	0	0
Average:	10,000,000,000	10,000
Upper limit:	10,000,000,000,000,000	10,000,000,000

This shows that in a fairly typical instance the stock produced by average environmental conditions is located very asymmetrically with respect to the two extreme limits, on this arithmetic scale. On a logarithmic scale the average production becomes much more nearly central, if a minimum of 1 fish is assumed:

	Log No. of fish
Lower limit:	0
Average:	4
Upper limit:	10

From this and other considerations it appears likely that our symmetrically distributed environmental variations will tend to act in relative rather than absolute fashion; so that, for example, if a given negative deviation produces a decrease in reproduction to half of the average level, the same positive deviation will increase reproduction to twice the average level. In other words, the figures representing environmentally caused deviations from the reproductive norm must be multiplicative rather than additive. To put them into this form, unity is added to each, so that deviations indicating below-average conditions (the *d*-items above) become divisors, those indicating above-average conditions become multipliers, and zero deviation is multiplication or division by unity. The result is shown in Table II. The absolute magnitude of the items shown is of course arbitrary, and it can be varied at will by multiplying the original random series (before unity was added) by any desired integer or fraction.

TABLE II. Series of numbers selected randomly from a population having the positive half of a normal frequency distribution with a standard deviation of unity, each number augmented by unity, and the series divided randomly into multipliers and divisors (the latter indicated by the suffix "*d*").

2.8*d*	1.1*d*	3.0*d*	2.2	1.9*d*	1.3*d*
1.5	1.1*d*	1.9	1.1*d*	1.0	1.6*d*
1.2	1.9*d*	1.4	1.0	2.0*d*	1.2
1.1	2.0*d*	1.1	1.4	1.1*d*	2.2*d*
1.2*d*	2.4	2.4*d*	1.7*d*	2.8	1.3*d*
1.0	1.5*d*	1.7*d*	1.7*d*	2.7	2.3*d*
2.0*d*	2.2	1.3	1.6	2.8*d*	1.2
1.1	1.3*d*	2.7*d*	1.8	3.5*d*	1.3
2.3*d*	1.2	1.5*d*	1.4*d*	1.2*d*	3.0
1.8*d*	1.9	1.9	1.0	1.2*d*	1.3
1.1*d*	2.0*d*	1.9*d*	1.8	1.7	2.2*d*
1.7	2.1*d*	2.1*d*	1.0	1.9*d*	3.2*d*
1.8*d*	1.5	1.8*d*	2.2	2.6	1.4*d*
2.5	1.3*d*	1.8	1.0	2.4*d*	1.5
2.0	1.6*d*	2.6*d*	1.5*d*	2.5	1.9*d*
2.0*d*	1.5*d*	2.3	2.2*d*	2.0*d*	1.1*d*
2.0	1.0*d*	1.7*d*	1.6*d*	2.2*d*	2.6
2.2*d*	1.9*d*	1.1	1.9	2.1*d*	2.5
2.2	1.9*d*	2.8	1.7	1.9	2.1*d*
1.3	1.3	1.3	2.6	1.8*d*	2.6*d*
1.4	2.1*d*	1.1*d*	2.4*d*	2.8	2.1*d*
1.2*d*	2.1	1.6	1.5	1.3	1.8
2.0*d*	1.2	1.2*d*	2.0*d*	1.7*d*	2.6*d*
1.7*d*	1.8	1.0*d*	1.7*d*	1.6	1.3
1.1*d*	2.0	1.7	1.0	1.7*d*	2.5*d*
1.5	1.1*d*	1.4	1.2*d*	1.8	2.7*d*
2.2	2.3	1.1	1.5	1.0	1.1
1.6	1.2	2.1*d*	2.0	2.0*d*	1.3
1.4*d*	2.0*d*	2.1	1.3*d*	2.0	1.7*d*
2.0*d*	1.6*d*	1.9*d*	1.8*d*	1.6	1.1*d*
1.3*d*	2.4*d*	1.2*d*	2.3	1.0*d*	2.0
3.4	1.1*d*	1.1*d*	1.6	1.3*d*	1.6
1.3	1.7*d*	1.6	1.3*d*	3.7*d*	3.3*d*
3.2*d*	1.9*d*	1.5*d*	1.7*d*	1.3	2.3*d*
1.3	2.7*d*	1.3	1.3*d*	1.0	1.5*d*
2.2*d*	1.5	2.2*d*	1.7	1.3	1.6*d*

SINGLE-AGE SPAWNERS

Since the figures of Table II constitute a model of variation in success of reproduction under the influence of density-independent factors alone, a picture of corresponding population fluctuations is obtained by multiplying some initial stock density by each item in succession. This is done in the lower line of

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Figure 11A; the upper line is from a different random series. The lines fluctuate a good deal and at times diverge considerably from the original abundance: at the end of the 70 generations shown in Figure 11 the Table II line has changed by a factor of 260; while if the same is continued through the whole 216 generations of the Table the net change is 5.5×10^{-7}, representing a relative decrease

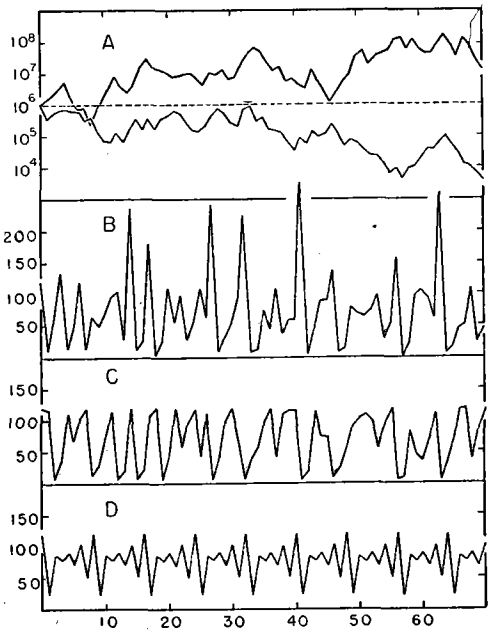


FIGURE 11. Fluctuations in ideal populations of single-age spawners. *Abscissa*—generations; *ordinate*—number of units of stock.

A. Two series of changes in a population from an initial density of 1,000,000, when it varies under the influence of density-independent factors only. The lower line corresponds to the first 70 entries of Table II, multiplied in sequence.

B. Changes in a population from an initial abundance of 119, as determined by the random factors of Table II and the reproduction curve of Figure 7, when the action of the latter precedes the former.

C. As in B, but with the random factors acting before the compensatory mortality shown by the reproduction curve.

D. Population cycle generated by Figure 7 alone.

from 55,000,000 to 1. This large change occurs in spite of the fact that Table II is constructed in such a way that, if it were continued over any really long period, the population should be above its initial level about as often and as much as it is below it[5]. The net change results partly from an excess of *d*-items in Table II, and partly from the fact that the *d*'s happen to be larger numbers, on the whole.

[5]A series of 1,000 "generations" constructed in a different but analogous fashion by Hutchinson and Deevey (1949, fig. 4) shows similar major long-period trends.

By chance the change indicated by Table II is considerably greater than would be expected to occur often in a series of only 216 generations. However it is easy to show that as the number of generations increases, the most probable divergence of population size from the original level also increases continuously. When the number of generations approaches anything appropriate to a geological time scale the likely change is very great indeed. For example, after 10,000 generations the standard deviation of the number of increases (or decreases), from the most probable number 5,000, is $\sqrt{10{,}000 \times \frac{1}{2} \times \frac{1}{2}} = 50$. This means that there is about 1 chance in 3 that the excess of increases or decreases will be 50 or more at that time, while the chances of the excess having been 50 or more at some time *during* the 10,000 generations are much greater. With an average change factor of 1.67 from one generation to the next (as in Table II), this means that the population is fairly certain to have been increased or decreased by a factor of about 1.67^{50}, or 140,000,000,000, somewhere along the line. Without wasting thought on the impossibility of a population of any sizable organism *increasing* this much, such a *decrease* would obviously bring even the most numerous fish stock, for example, to extinction. Thus no population of single-age spawners can survive if its reproduction is completely at the mercy of density-independent influences in the environment. In the same way a gambler will in the long run lose everything if his opponent has equal skill and unlimited resources. (In practice, of course, the effects of random environmental factors would usually vary as between different parts of the animal's range, and local extinction could be followed by recolonization from adjacent areas.)

The average interval between peaks of Table II should be close to the 3 which is characteristic of a random series; in actuality it was 3.09 when ties were randomly divided into higher and lower values. However this is not what determines the period of a series such as Figure 11A. In the latter a peak occurs whenever a multiplier is followed by a divisor, and the average interval between peaks is nearly 4 years, again adjusting for ties. The theoretical or long-term average period can be determined as follows: since a peak or trough occurs whenever there is a change from multiplier to divisor or divisor to multiplier, the expectation that the first such change will occur between adjacent cells is one-half, the expectation that it will occur between a cell and the next but one is one-fourth, that it will occur at the third cell is one-eighth, and so on. The average of these intervals 1, 2, 3, etc., weighted as to frequency, is exactly 2, which represents the average peak-to-trough interval. Since peaks and troughs are equally common, the average peak-to-peak or trough-to-trough interval is therefore 4.

However it must be emphasized that this 4 represents the average interval between *all* peaks, regardless of size. Casual inspection of Figure 11A might give a different impression. The eye tends to ignore the smaller humps, and to impose a certain regularity among the rest by magnifying those of intermediate size when they happen to fall into a sequence with large ones, and diminishing them when they do not. In this manner the lower line could "suggest" a cycle of 11 or 12 years, with 4 peaks and 4 troughs actually showing.

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MULTIPLE-AGE SPAWNERS

To obtain a model of the consequences of random fluctuations upon reproduction of multiple-age stocks, the random values of Table II have been applied to a "population" constructed on the same basis as Figure 9—that in the spawning stock there are 4 ages, that each year-class produces eggs in the ratio 2:3:3:2 in the 4 years of its reproductive activity, and that each fish spawns for the first time 4 years after it was itself a fertilized egg. The resulting curve is shown in Figure 12A (it starts from an age distribution characteristic of the reproduction-curve cycle of 12D).

The most distinctive characteristic of the line of Figure 12A is that it tends to rise. This increase occurs in spite of the fact that divisors happen to be in excess in Table II, and if the series is continued a little farther much higher values are encountered. The fact is that, if continued, the population will increase without limit. The reason is that the contributions of the several year-classes to a given year's spawning must be added arithmetically, while expectation of

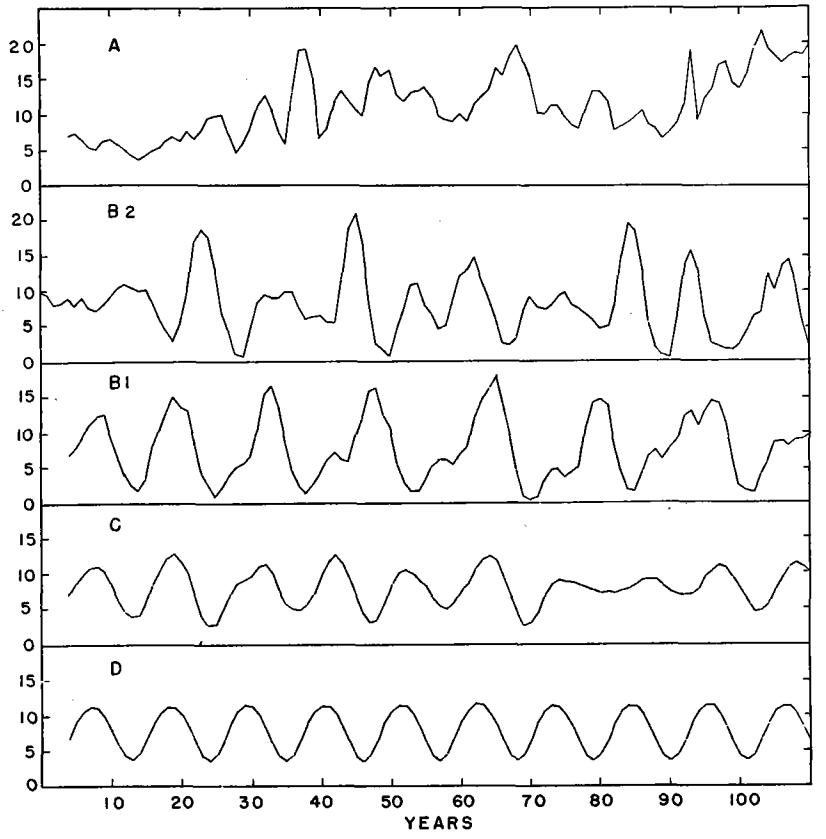


FIGURE 12. Fluctuations in a population of multiple-age spawners, as determined by the random series of Table I (curve A), by the reproduction curve of Figure 7 (curve D), and by combinations of these (curves B and C). Axes as in Figure 9.

survival of an egg is proportional to random environmental effects having a symmetrical distribution so constituted that any increase is greater, absolutely though not relatively, than the corresponding decrease. It would be difficult, of course, to prove that environmental effects upon reproduction are actually distributed in just this manner. On the other hand it was shown above that they are very unlikely to be merely additive; and any intermediate condition will result in a slower but still unlimited upward trend to a graph such as Figure 12A.

It thus appears that any likely random sequence of environmental changes, even one which for single-age spawners causes a catastrophic decline, can be made to produce instead a large increase in abundance of stock by the simple device of dividing the spawning of each year-class among several calendar years. Even though wholly non-compensatory reproduction cannot occur in nature, this circumstance may have considerable advantage for any population in which random variability is large and/or the left limb of the reproduction curve not very steep (e.g., as in Figs. 1, 2a, 3 or 4). Indeed it is easy to show that in the presence of random variability a multiple-age spawning stock can maintain itself even when its reproduction curve lies wholly below the 45-degree line—though of course it cannot be *too far* below. Thus there is a sound ecological basis for the customary occurrence, in nature, of reproductive assemblages consisting of more than one age-group.

As regards period, we find again that the peaks and troughs of Figure 12A have a certain apparent regularity in spite of their random origin. Discounting the first 15 years as being still somewhat under the influence of the reproduction-curve cycle, and counting all peaks no matter how small, there are 17 peak-to-peak intervals in the figure, of which no less than 10 are of 5 years' duration—the average interval is 5.2 and the range 2 to 7. The fact that the average interval is longer than the 3 which is characteristic of a simple random series (and longer than in the series for single-age spawners) is explained by the method used to obtain it. This is quite similar to taking a "running average" of four items at a time, and Cole (1951, fig. 1) has shown that applying such a procedure to a random series increases the mean peak-to-peak interval. "Artificial" trends of really long period are not as apparent in Figure 12A as in 11A, but a longer series might have confirmed them.

In summary, the characteristics of population changes which would result from density-independent factors acting alone (if that were possible) are as follows:

1. A population of single-age spawners would vary widely above and below its initial abundance; eventually, after a few thousand generations at most, it would either become extinct or, more likely, be fragmented temporarily into small independent units.

2. Populations of multiple-age spawners would increase in abundance indefinitely, though not without ups and downs.

3. The average peak-to-peak period for lines of single-age spawners would be 4.

4. The average peak-to-peak period for multiple-age spawners would be

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more than 3, and would be the greater, the greater was the number of ages in the spawning stock; in the example used it was about 5 years.

5. For single-age spawners, and probably for multiple-age spawners as well, "cycles" having longer periods than these would tend to be apparent in graphs of abundance, because of conscious or unconscious mental suppression of small peaks and troughs, and regularization of large ones.

COMBINATIONS OF COMPENSATORY AND NON-COMPENSATORY MORTALITY

In natural populations non-compensatory mortality is superimposed upon the reproduction expected from density-dependent factors, and the curve of population fluctuation is the resultant of both. The random effect can be introduced either before or after the amount of reproduction indicated by the curve is written down, according as the random factors are thought to act before, or after, the compensatory ones. The latter procedure gives greater influence to the random element, and may correspond better to events in nature, though the two types of mortality may of course act concurrently, and to a considerable extent probably do.

Single-Age Spawners

Combinations of Table II or similar series can be made with the various kinds of reproduction curve shown in Figures 1–8. It seems unnecessary to reproduce examples of most of these. With a flat-topped curve such as Figure 2b, and the scale of random variability shown in Table II, the resultant reproduction is practically always equal to the product of the equilibrium abundance and the multiplier or divisor for the year in question. Only an improbably large deviation could shift the population over to the ascending part of the curve for a year. Relatively slight modifications of this situation are obtained when broad-domed curves having a stable equilibrium point are used (e.g., Figs. 3 or 4); the principal difference being that, on the whole, reproduction is somewhat less.

Fluctuations produced by a combination of the reproduction curve of Figure 7 with the random series of Table II have some of the characteristics of either component (Fig. 11). The regular series A–H of Figure 7, repeating at 8-year intervals, is replaced by an irregular one. The average peak-to-peak period is about 3.5—slightly less than that of the random series (which was close to 4) and greater than the 2 of the reproduction-curve sequence. The average amplitude of the "cycles" (ratio of peak to preceding trough) is greater than what either component series exhibits, but there are of course no extreme trends in abundance such as resulted from random causes alone. There is a pronounced tendency for each peak to be followed immediately by a trough, which is a peculiarity also of the random-curve series.

The above description applies whether the random influence operates before the reproduction-curve (Fig. 11C) or after it (Fig. 11B). The two kinds of series are quite similar, but the latter of course has the greater amplitude of changes.

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Multiple-Age Spawners

Combination of compensatory and non-compensatory reproduction in multiple-age stocks will be considered only for the case where the former precedes the latter, and will be illustrated by means of a population in which contributions to reproduction are spread over four years in the ratio 2:3:3:2, as in Figure 9.

In general, the kind of fluctuation which results from any combination of compensatory and non-compensatory mortality factors depends upon the relative magnitudes of the two components. The action of the same series of random factors is shown in Figures 12B and 12C at two levels of intensity which are in the ratio of 5:1; Table II is used for B, and values one-fifth as large for C, i.e., 1.36*d*, 1.10, 1.04, etc. In each case they are combined with the reproduction indicated by Figure 7. The same initial age distribution is used for all series of Figure 12.

By itself, Figure 7 yields the steady oscillation of Figure 12D. At the lower intensity of random effects (line C) the population cycle determined by the reproduction curve is not too seriously altered: there is variation in amplitude, and the peaks move out of phase by as much as a fourth at times, yet the prevailing periodicity can be determined fairly accurately from even a short series. At the higher intensity of random effects, however, much greater disturbance is evident; it is followed through in two lines of Figure 12, beginning with B1 and continuing in B2. As regards *period*, the 11-year cycle is first increased to 15–17 years, then reduced to 7 or 8 (counting major peaks only). Though the average over a long period may thus tend to 11 years, peaks and troughs move out of and back into phase with the reproduction-curve cycle and also with the random "cycle", which are its determiners. More serious is the fact that the smaller peaks and troughs of what appears to be the main series are in some cases impossible to distinguish from the "artifacts" resulting from random fluctuation: hence the average 11-year period above could not be discovered in practice, even with a very long record. As regards *amplitude*, the maximum ratio of peak to adjacent trough is much greater than is found in either the reproduction-curve cycle or the random series.

When the random element is given still greater relative importance, the reproduction-curve element becomes unidentifiable as such. However the latter continues to make an important contribution to the resulting population changes. It makes peaks and troughs much less numerous than they would otherwise be, and it provides a control of limits of abundance—that is, the progressive upward tendency of the random curve is effectively curbed.

The relative importance of random and compensatory factors in determining population abundance also depends somewhat upon the number of ages represented in the spawning stock. When this is large, random factors are less effective in disturbing reproduction-curve cycles.

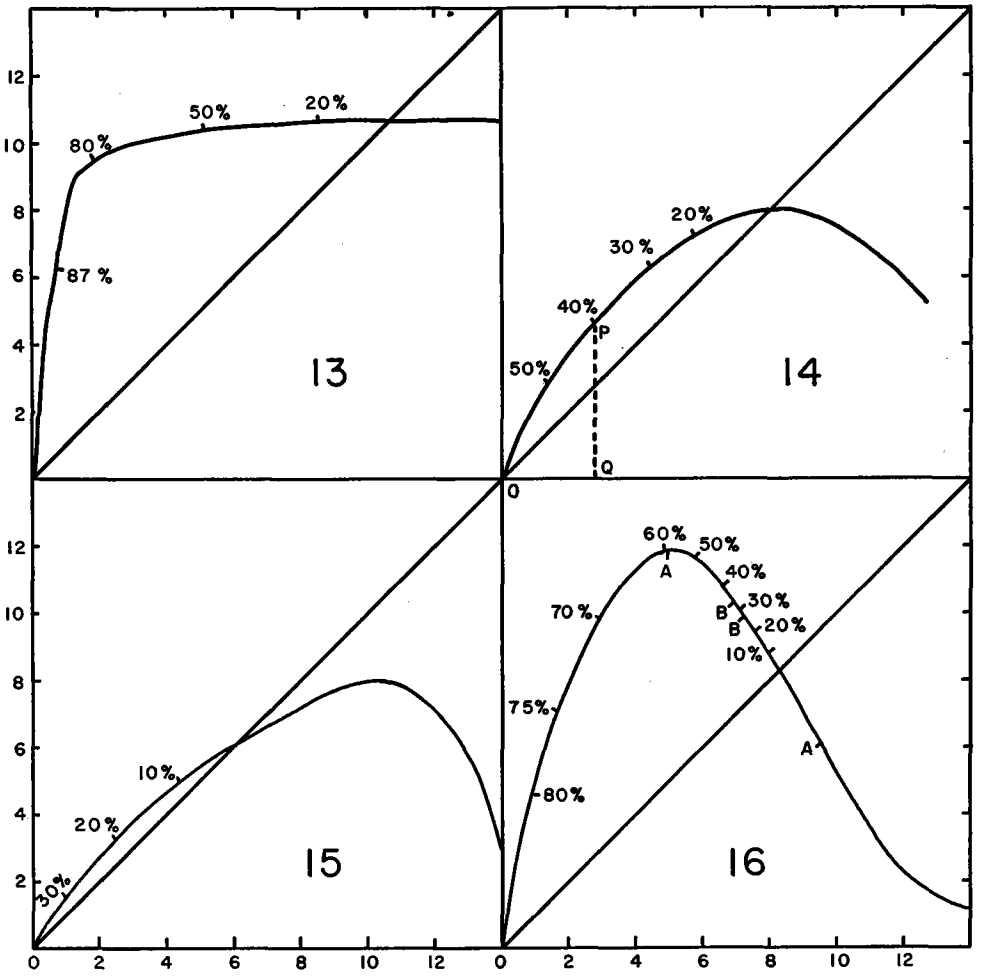
Effects of Removal of Mature Stock

Single-Age Spawning Stocks

Many natural populations have a part of their members removed by man, either because they are useful to him or, in the case of harmful species, because

he hopes to reduce their abundance. The case where only *mature* stock is taken by the fishery is considered in examples to follow. Many fisheries take a portion of the immature stock also, but this appears not to add any new principle to what is considered below, provided the region of compensatory mortality is not invaded.

The effect upon the various population parameters of removing a constant percentage of the mature stock, before reproduction, is shown for four types of single-age-group populations in Figures 13–16. In Figure 14, for example, a spawning stock OQ produces filial spawners equal to PQ. However, after the fishery has removed 40 per cent of the mature stock the number remaining is equal to OQ, which again produces PQ, and so on. Thus P is the equilibrium position for 40 per cent



FIGURES 13–16. Equilibrium densities of stock for the various rates of exploitation indicated by percentages on the reproduction curve. Stock density before fishing is the ordinate value corresponding to the point on the curve opposite the percentage exploitation in question. Axes as in Figures 1–4.

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exploitation. The fraction of the mature stock which must be removed prior to spawning, in order to maintain equilibrium at any point on the reproduction curve, is equal to the complement of the reciprocal of the slope of a line joining that point to the origin.

The general effect of exploitation is to move the point of equilibrium abundance to the left on the reproduction curve. In Figures 13–15 this means that under exploitation the equilibrium population is always smaller than it is under natural conditions. In Figure 16 (which is the same as 7), removal of part of the spawning stock at first results in a *larger* equilibrium level of stock, and the same is true of Figures 5, 6 and 8. When rate of exploitation becomes large enough, however, the stock is again reduced. Equilibrium points for various rates of exploitation are indicated on Figures 13–16.

Oscillating equilibria are much changed by exploitation. Their amplitude is reduced, and the number of positions through which they swing is decreased, or oscillation may be eliminated entirely. However, stable oscillations persist when exploitation is light to moderate. In Figure 16, for example, with 10 per cent removal the stock traces the cycle 11.9, 4.0, 10.9, 5.5, 11.9, etc.; with 20 per cent removal it alternates between the levels of 6.1 and 11.9 units (A–A); with 30 per cent removal the oscillation is reduced to between 9.9 and 10.3 (B–B), so that it would not be distinguishable in practice; and with 60 per cent removal it disappears completely[6].

The point of maximum sustained yield can easily be computed from graphs like Figures 13–16. For example, in Figure 16 it is at 65 per cent exploitation; in Figure 13 it is close to 80 per cent; in 15 only about 18 per cent. In general, the greater the area between the reproduction curve and the 45-degree line above the latter, the greater is the optimum rate of exploitation.

Beyond the level of maximum yield lies a level of maximum permissible exploitation above which the stock is progressively reduced to zero. This limit is a function of the maximum angle made by any line joining a point on the reproduction curve to the origin; and as noted above, the maximum rate of removal which can be sustained is equal to the complement of the cotangent of that angle.

If a man's interest in a single-age population, of a noxious insect for example, is to reduce it to as low a level as possible by increasingly intensive destruction, he can be sure of making *direct* progress toward the desired goal only if the reproduction curve is one of the types shown in Figures 1 and 2. If it is like Figures 3 or 4, and stock happened to be on the outer limb, some moderate rate of removal would dampen a crash which otherwise was imminent. With curves of types 5–8 a paradoxical situation develops. Moderate destruction of adults will in general tend to stabilize the population at or about some magnitude *greater* than its primitive average abundance. In Figure 16 maximum abundance is con-

[6]When equilibrium is unstable the *average* size of the populations of the cycle tends to be less than the equilibrium level. In Figure 16 the average is 7.7 for no exploitation, 7.8 for 10 per cent, and 9.0 for 20 per cent, as compared with the equilibrium values of 8.2, 8.6 and 9.4. For 30 per cent or greater exploitation, equilibrium stock and average stock are the same.

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sistently achieved when as much as 60 per cent of the spawning stock is destroyed each year. To reduce the population below the primitive average, more than 73 per cent must be destroyed. Thus although sufficiently intensive effort will be successful, moderate destruction of the mature population is worse than no action at all[7]. Furthermore, if an intensive campaign of continuing control of a variable species is decided upon, it is most efficient to begin it at a time that the population is at a low point of its cycle, i.e., when the pest may be doing no particular damage.

Multiple-Age Spawning Stocks

The effect of a fishery upon stocks containing more than one age-group of spawners parallels what has just been found for single-age-group stocks. If the stock is one for which a stable equilibrium exists (Figs. 1–5), adding a fishery does not disturb the stability, but the abundance of the population is changed (in 2b the change begins only after the inflection point is reached). With the curves of Figures 1–4 the change is in the direction of a decrease in abundance, but with Figure 5 exploitation at a moderate rate *increases* the adult stock, until the dome of the reproduction curve is reached.

Stocks which perform regular oscillations in the absence of a fishery (Figs. 7 and 8) are changed in three respects when exploitation begins: (1) their average equilibrium abundance is at first increased, but later decreases again if exploitation becomes sufficiently intensive; (2) the amplitude of oscillation decreases; and (3) the period of oscillation tends to decrease slightly.

1. The first-named effect can be estimated rather easily, since the average abundance of multiple-age stocks proves to be practically the same as the equilibrium abundance. The latter has been calculated for Figure 16 using various rates of exploitation, as shown on that Figure and in Table III. The maximum abundance is at the dome of the curve, and maximum catch is obtained slightly to the left of it, at 65 per cent exploitation.

2. The advent of fishing not only affects the total abundance of a stock, but also gives it a younger age composition, because survival rate from year to year is reduced. In the example of Table IV, based upon Figure 16, the fish are assumed to mature first at age III. Fishing takes place just prior to spawning and rate of fishing is the same for all ages. Natural mortality occurs between successive fishing-and-spawning seasons, and is 20 per cent from age III to age IV, 20 per cent from IV to V, 30 per cent from V to VI, and 100 per cent after the spawning at age VI. The average weight of a fish at time of fishing-and-spawning is 2 units at age III, 4.28 units at IV, 7.14 at V and 9.52 at VI; and egg production is proportional to weight. Under these conditions the *equilibrium* distribution of the contributions of the several age-groups to egg production, at different rates of exploitation, is shown in Table IV.

[7]A similar conclusion is reached in Nicholson and Bailey's (1935) detailed analysis of host-parasite interaction in insects. Varley (1947) concluded that a trypetid fly population was maintained at about 10 times the density it would otherwise have had, because of the presence of non-compensatory causes of mortality which killed host and parasite equally.

TABLE III. Average or equilibrium abundance of stock, by weight, in a stock having more than one age at maturity, and the catches at various rates of exploitation; from Figure 16.

Rate of exploitation	Stock before fishing	Catch
%		
0	8.2	0
10	8.8	0.9
20	9.5	1.9
30	10.1	3.0
40	10.8	4.3
50	11.5	5.8
60	11.9	7.1
65	11.4	7.4
70	9.7	6.8
75	7.2	5.4
80	5.0	4.0

TABLE IV. Relative weights and contributions to reproduction of the age-groups in the stock described on page 583.

Rate of exploitation	Fraction of eggs contributed by age			
	III	IV	V	VI
%				
0	0.200	0.300	0.300	0.200
10	0.233	0.314	0.283	0.170
20	0.273	0.327	0.261	0.139
30	0.320	0.336	0.235	0.109
40	0.377	0.339	0.204	0.081
50	0.446	0.333	0.165	0.055
60	0.526	0.315	0.127	0.032
70	0.619	0.278	0.083	0.020
80	0.737	0.221	0.042	0.000

The first line of Table IV was arranged to have the same distribution of contributions to spawning as used for Figure 9F. In Figure 17 are shown cycles obtained from the 10 per cent and 20 per cent lines of Table IV, starting from the equilibrium distribution characteristic of no fishery[8]. At 10 per cent exploitation the oscillation of the population is maintained, though at reduced amplitude and about a higher mean level. At 20 per cent exploitation oscillation gradually decreases to an inconsiderable amplitude, and at higher rates of exploitation it

[8]This procedure is the only practicable one, but it ignores the fact that the age distributions of Table IV would themselves be completely realized only after a period of 4 years from the beginning of fishing. Consequently the equilibrium situations in Figure 17 would actually be approached more gradually than the figure indicates.

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disappears completely. However even at 20 to 40 per cent exploitation the oscillation persists through several cycles.

3. The length of a cycle was found earlier to be twice the mean length of a generation (interval from egg to egg). In the first line of Table IV this is 4½ years, and the cycle is 9 years. With the higher exploitations of Table IV a shift of age distribution toward younger fish occurs, and we accordingly expect

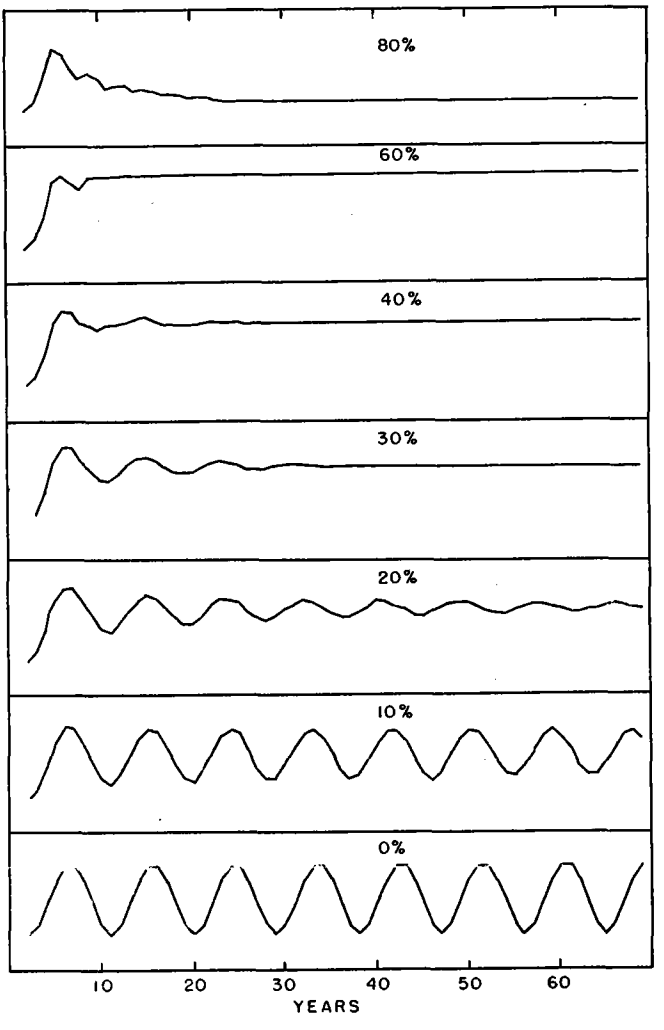


FIGURE 17. Reactions of an hypothetical population to exploitation. The population is one regulated by the reproduction curve of Figure 7, with ages III, IV, V and VI contributing to egg production in the ratio 2:3:3:2 when there is no exploitation. The stable cycle characteristic of no exploitation is shown by the 0% line. The other lines show the effect of annual removal of the indicated percentage of the mature stock from a population which initially had the age distribution characteristic of the start of the ascending phase of the no-exploitation cycle. (See also footnote 8.)

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a shorter cycle to appear. In this example, however, the cycles disappear before much decrease in period can take place. At 10 per cent exploitation average age of contribution to reproduction is reduced to only 4.39, and at 20 per cent it is 4.27. The corresponding predicted cycle lengths are 8.8 and 8.5 years, and the peaks shown in Figure 17 do in fact become progressively a little closer together. In a somewhat more realistic example there would have been more than four age-groups in the stock under conditions of light exploitation or none, and in that event the reduction in period of oscillation with increased fishing would be greater. However it is doubtful whether in nature the reduction would ever be great enough to be identifiable with certainty, before increasing exploitation removed oscillation of this type entirely.

When the reproduction curve is dome-shaped, *control* of an undesirable population of multiple-age spawners presents some of the same difficulties as described earlier for single-age stocks. However the initial favourable reduction in abundance will tend to last longer—i.e., until the increased broods of young can grow to the size at which they cause damage. Only when this first phase is over, and the fishing effort has to cope with the larger broods which come from the reduced spawning populations, will it be apparent whether or not control can be permanently effective. Few recorded attempts to reduce nuisance fishes have lasted beyond the initial stage of removal of old stock. Foerster and Ricker (1941) have described the rather easy success achieved in this preliminary phase of control of squawfish (*Ptychocheilus*) in a small lake. The experiment was subsequently continued until one or more large broods of young began to grow into the damaging size range, but it did not last long enough to see if these could be kept sufficiently reduced by appropriate effort. In Lesser Slave Lake, Alberta, unlimited fishing for ciscoes began in 1941 with the hope of thus reducing the lake's population of the tapeworm *Triaenophorus*, and as a result the spawning population has been changed gradually from one 5–7 years old to one now (1951) mostly 2 years old. However recruitment and rate of growth have increased sufficiently to maintain a high level of catch and a fairly large population mass, so that control is not yet effective for the purpose intended (Miller, 1950, 1952).

When non-compensatory variability is added to the effects of exploitation and the reproduction curve, the kind of result obtained differs little from what was discovered from Figure 11. The amount of disturbance introduced depends of course upon the relative magnitude of the random factors.

EXAMPLES OF REPRODUCTION CURVES

A wider recognition of the different possible kinds of reproduction curves, and their different properties, will be of value mainly as a guide to current and future research. Such curves should be useful not only as a possible clue to the nature of observed fluctuations, but still more in estimating expected average recruitment when stock is reduced (or increased!) by fishing it, at different intensities. Knowledge of the reproduction curve will thus complement information on rate of growth and natural mortality, and permit more accurate prediction and regulation of fish catches.

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Some examples are given below of reproduction curves, for fish and a few other animals, which can be plotted from available data. In several instances information is available concerning only the first part of the life history, but since that is where most compensatory mortality is expected, the data are included. A disadvantage of such curves, however, is that the "45-degree line" can usually be located only approximately, if at all.

PACIFIC HERRING (*Clupea pallasi*)

The herring of the west coast of Vancouver Island are a natural unit which has been given special study. Recently Tester (1948) examined the relation between the intensity of spawning in the brood year, and an estimate of resulting year-class strength based on catch per unit effort in the various years that each brood contributes to the fishery. Tester had two indices of spawning; using "Index A", which was available for five years longer than "Index B", he computed an inverse correlation of —0.30 between spawn deposited and resulting year-class strength for the period 1931–43, though this figure does not differ significantly from zero.

Mr. J. C. Stevenson has very kindly added data for more recent years, and these are plotted in Figure 18. In this figure the more elaborate "Index B" is used, but the five years 1931–34 are included (open circles) using values from the regression of B on A for the years of overlap. The negative correlation —0.16, for the years 1931–49, differs from zero even less than the one computed by Tester, principally because a series of abundant year-classes from 1943 to 1947 has increased the range of variability. However it is still true that the largest

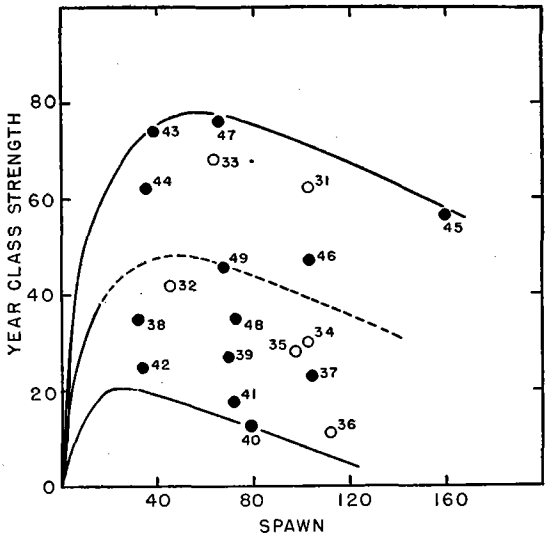


FIGURE 18. Relative strength of year-classes of a Pacific herring stock (ordinate) plotted against relative amount of spawn deposited in the brood year (abscissa). The two solid lines include the observed limits of variation, while the dotted one is an approximate middle value. Data from Tester (1948) and Stevenson (MS).

broods have been produced by smaller-than-average spawnings, whereas the smallest broods have been produced by larger-than-average spawnings.

Dr. Tester and others have also considered the possibility that trends in herring recruitment may exist corresponding to some as yet unidentified rhythm of the ocean environment; and the 1943–47 stanza of good year-classes is in fact fairly well set off from earlier (1934–42) and later (1948–49) stanzas of lower production (Fig. 18). If there is really any systematic effect here, one may notice that *within* each of the two longer stanzas a negative correlation between spawning and recruitment is indicated.

Pink Salmon (*Oncorhynchus gorbuscha*)

Among pink salmon there are two completely separate reproductive lines[9], so that is is very easy to identify the progeny of any year's spawning when it matures two years later. From Pritchard's (1948a, b) summary of the reproduction of the stock which spawns in McClinton Creek, Graham Island, British Columbia,

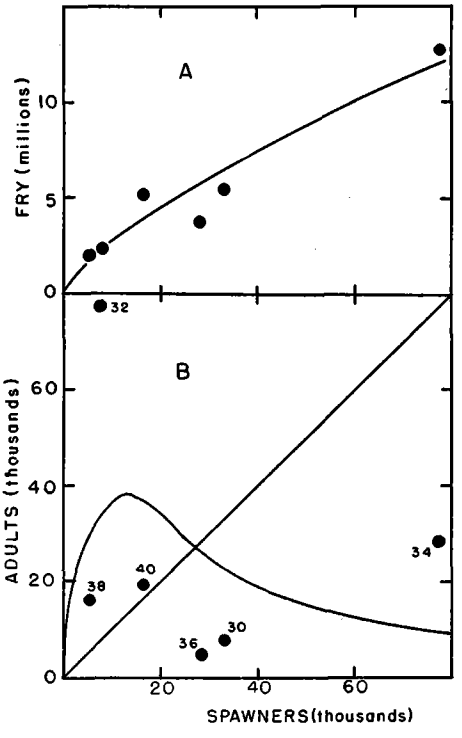


Figure 19. A—Fry produced in millions; and B—returning adults in thousands; plotted against the number of pink salmon spawners in the brood years indicated, at McClinton Creek, B.C. Data from Pritchard (1948a, b).

[9]Separate or nearly separate breeding stocks of salmon are commonly called "cycles", "years" or even "races" in western North America. The term "lines" was proposed by Huntsman (1931), and is used here because it avoids confusion with other kinds of cycles, etc. (cf. Clemens, 1952).

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the reproduction curve of Figure 19B is plotted. The progeny shown does not include the commercial catch, which is a considerable but unknown quantity, and hence all points would be moved upward on a graph that represented total adult production of the spawnings indicated. No very clear idea of the shape of the reproduction curve can be obtained, but it seems to slope downward from 15–20 thousand spawners.

Dr. Pritchard also had records of fry production, and these are plotted in Figure 19A. The fact that *percentage* fry production decreases as population increases indicates that compensatory mortality occurs in the freshwater part of the life history. That compensation occurs also during saltwater life is suggested by the change from a generally upward slope in Figure 19A to a generally downward slope in 19B. Obviously too there is much random variation in mortality at both stages, but particularly in the ocean. The ocean phase includes the effect of exploitation by the fishery, which is certainly variable and might tend to be

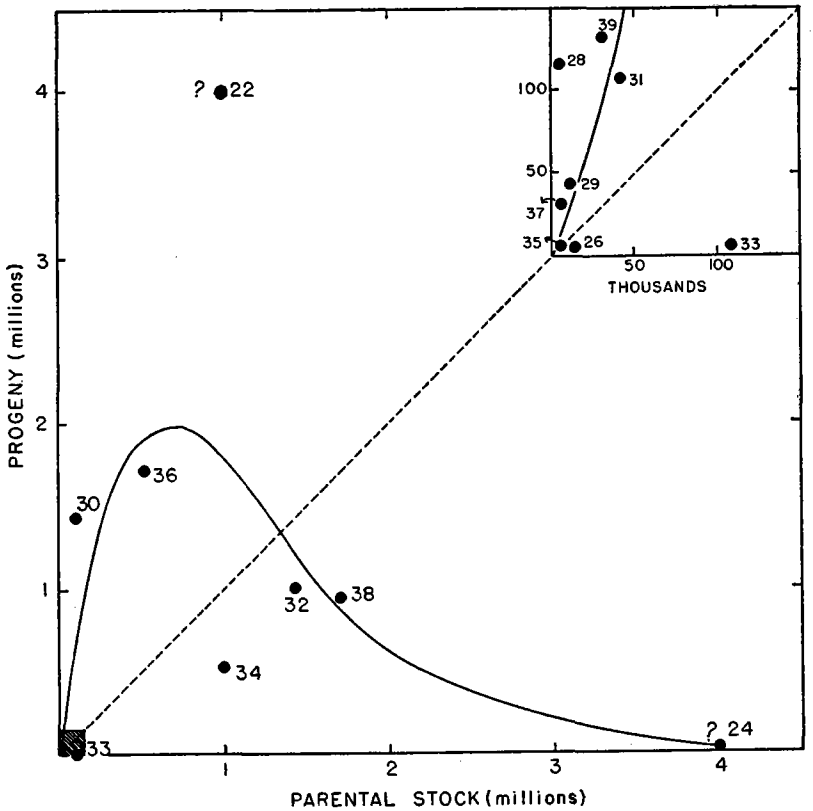


FIGURE 20. Reproduction curve for pink salmon of the Karluk River, Alaska, based upon fence counts except in 1922 and 1924 (see text). The shaded square in lower left is shown on a larger scale in the upper right corner. Commercial catches are not included in the progeny. (From unpublished data of the United States Fish and Wildlife Service, Pacific Salmon Investigations.)

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compensatory, because more seiners are attracted to the bay off McClinton Creek when fish are numerous there.

A reproduction curve having a very wide range of parent-stock densities is that for pink salmon of the Karluk River, Alaska. This is shown in Figure 20, plotted from data supplied by Messrs. C. E. Atkinson and C. J. Burner of the U. S. Fish and Wildlife Service. "Progeny" figures do not include the commercial catch; adding the latter could not alter the general form of the curve, but would raise it farther above the 45-degree line. The 1926–40 data are from complete or nearly-complete fence counts. The 1924 escapement was estimated as "over 4,000,000" (Barnaby, 1944, p. 257). Unfortunately there is no numerical estimate of the 1922 run; however, since it is called "large" on the same page as the 1924 run is called "tremendous" (Barnaby, p. 249), it might be assigned a figure of about one million, and on that basis a point for 1922 is plotted in order to better indicate the vertical range of reproduction.

Coho Salmon (*Oncorhynchus kisutch*)

The experiment at Minter Creek, near Tacoma, Washington, has provided information on the relation of spawning stock to smolt output (Smoker, 1954). Production of year-old smolts is related to number of female spawners in Figure

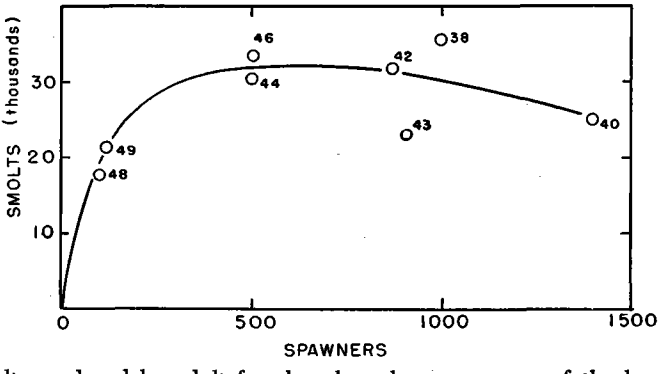


Figure 21. Smolts produced by adult female coho salmon spawners of the brood years indicated, at Minter Creek, Washington. Data from Smoker (1954).

21, based on the same data as the upper half of Smoker's figure 29. The productive capacity of the stream appears to be 25–35 thousand smolts, depending mainly on volume of flow (Smoker's fig. 29, lower half), and if adjustment is made for the latter, the already-flat outer limb of Figure 21 becomes even flatter, with the deviations of the points from the line considerably reduced.

Estimates of total adult production from these smolts are not possible; there is no record of catch, and even the number of returning spawners is available for only four years.

Sockeye Salmon (*Oncorhynchus nerka*)

The sockeye of Karluk Lake, Alaska, mature at 3 to 8 years of age, the mean being a little over 5 years. Barnaby (1944, fig. 3) has plotted a graph of spawning

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stock against adult progeny produced; the latter was obtained by summing the contributions of the 1921–29 year-classes to subsequent catches and breeding stocks. This graph is practically equivalent to a reproduction curve based upon eggs, the only difference being that possible variation in the number of eggs produced by females of different ages is not taken into account.

Barnaby separates "spring" and "fall" fish in his plotting. Either in this form, or when the data for each year are combined (Fig. 22), there is much variability in production, but a peak apparently occurred when the total spawning stock (spring plus fall) was between 1 and 1½ million fish. Unfortunately, both before and after this period the sockeye stock of the lake decreased, so the 1921–29 results cannot apply either to the earlier period when *catches* of 2–3.5 million fish were customary, or to the modern period when the total stock has fallen below a million. The causes of this decrease are still under study.

An opportunity to document the *ascending* phase of a sockeye reproduction curve is afforded by the "late Shuswap" or Adams River race on the Fraser River. A blockade in 1913 reduced this and other Upper Fraser races of sockeye to a level that was only a small fraction of their natural abundance. These fish mature preponderantly at 4 years of age, so that an adequate reproduction curve

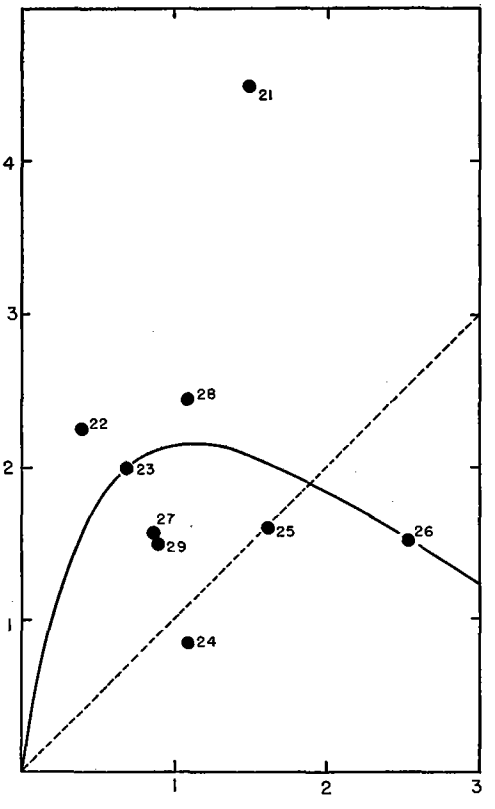


FIGURE 22. Number of mature sockeye (catch plus escapement) produced by the year-classes of 1921–29, at Karluk Lake, Alaska. Data from Barnaby (1944).

can be plotted using estimates of spawners and catches every fourth year (Fig. 23). Sizes of the spawning population are based on estimates of Fishery Inspectors and, in recent years, on figures given in Annual Reports of the International Pacific Salmon Fisheries Commission; the catch is taken as the total Fraser commercial catch each year, less 1,200,000 fish representing the approximate production of other races in this series of years. The curve has a much steeper ascending limb than the Karluk curve; there is a suggestion of a maximum at 1–1.5 million spawners, and possibly a start on the downward path beyond that point.

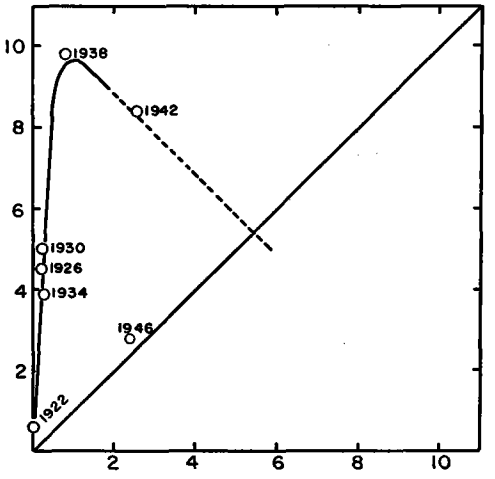


FIGURE 23. Reproduction curve for the late Shuswap race of sockeye salmon, based upon estimated numbers of spawners and an approximate division of the commercial pack. *Abscissa*—estimated number of spawners in the years indicated; *ordinate*—resulting catch plus escapement. Both scales are in millions of fish. The year of spawning is shown beside each point. From data of the B.C. Department of Fisheries and the International Pacific Salmon Fisheries Commission.

The steepness of the left limb of the Shuswap curve is worth noting, being of the same order as that of Figure 8. If the right limb has any considerable slope downward, it would have a tendency to mould chance fluctuations into swings back and forth across the 45-degree line. In this connexion one may note that the original dominant line of sockeye on the Fraser, of which the late Shuswap race was a component, is supposed to have been considerably less abundant in 1909 than in either 1905 or 1913 (Babcock, 1914). This might represent two swings across an equilibrium position, under the then conditions of a fairly low percentage exploitation of this line by the fishery (compare A–A of Fig. 16). Under present conditions the fishery normally takes enough to keep the spawning stock near the dome of the curve, and special measures are taken to ensure this much escapement when below-normal reproduction has occurred, as from the 1946 spawning (Royal, 1953).

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HADDOCK (*Melanogrammus aeglifinus*)

Herrington's (1941, etc.) analysis of the George's Bank population is particularly valuable because he was able to obtain from market reports an estimate of the smallest commercial size (scrod—mostly age III) independently of and in addition to the larger sizes. This yields an estimate of recruitment from year to year, which in turn is a reflection of success of reproduction. When size of spawning stock in the years 1912 to 1929 is plotted against scrod produced (solid circles, Fig. 24), a steep outer limb of a recruitment curve is apparent. The 45-degree line has been located very approximately by considering that a pound of scrod during those years had an expectation of egg production, throughout its life, equivalent to the actual egg production in one year of 10 pounds of mature haddock; and by assuming that scrod were half as vulnerable as were adult fish

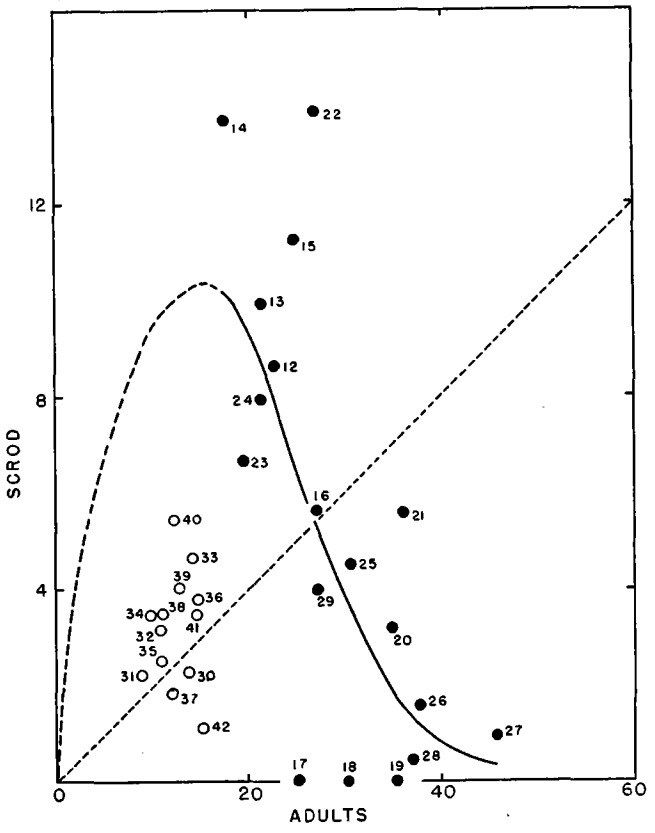


FIGURE 24. Abundance of adult haddock on George's Bank during spawning season (February–April), related to abundance of scrod (age III, mostly) at the same season three years later. Both scales are in terms of thousands of pounds per day caught by trawlers of a certain size class, the figures for adults being smoothed. Black circles represent the years 1912–29, open circles 1930–43. The reproduction curve is drawn for the earlier period only. The diagonal is an *approximate* 45-degree line for that period (see text). Data from Herrington (1948).

to the trawlers whose catches supply these data. Figure 24 is the same as Herrington's figure 11 of 1948, except that we have used catches of the February-April period only, for both scrod and adults. Herrington was of the opinion that the critical time for competition between adults and young, during those years, was when the young were about a year old (his fig. 12); however the two graphs are much the same and do not provide a basis for decision on this point. Either one supports Herrington's interpretation of the fluctuations in adult stock over the period in question, which consisted of two complete oscillations (Fig. 25). The trough-to-trough periods were 9 and 8 years, and trough:peak amplitude was about 1:2. The observed period implies a median age of 4 or 5 years for the production of eggs by the fish of each brood, and this seems consistent with what is known of the size- and probable age-structure of the haddock population in those years.

Thus Herrington has described the only example yet known in which oscillation of a multiple-spawning-age fish population might plausibly be ascribed to the simple effect of a steep reproduction curve. His account of the probable cause of the inverse stock-recruitment relationship emphasizes competition for food between adults and young. Intra-brood competition at an earlier age seems another possible factor, but the feeding habits of haddock are said to preclude significant cannibalism.

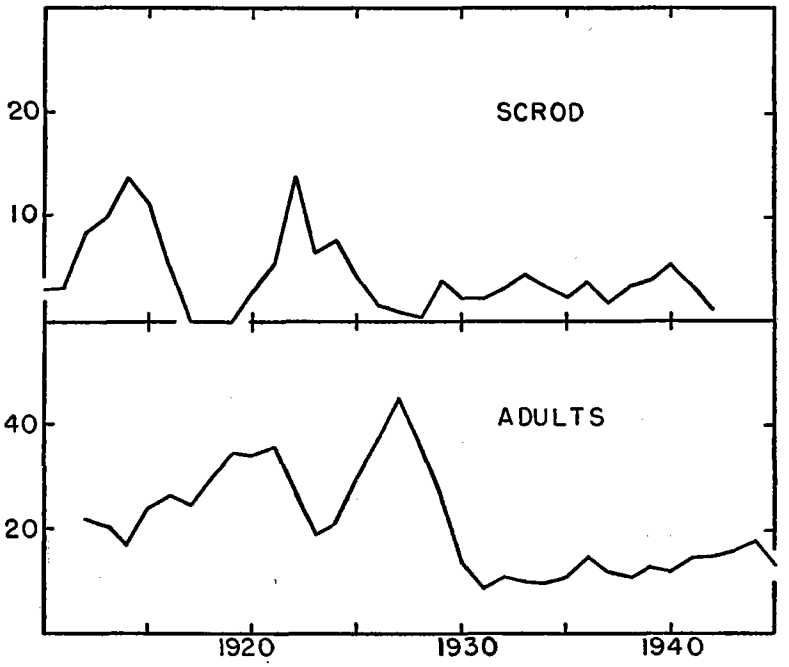


FIGURE 25. Catch per unit effort of adult haddock, related to that of the scrod (age-III fish) produced by them. The years shown on the abscissa are those in which the adult catch was made; the scrod on the same ordinate were taken three years later. From the same data as Figure 24.

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Points for the years 1930–43 are plotted as open circles in Figure 24. About 1930 the haddock fishery changed radically within a few years; total effort increased, and otter trawls to a large extent superseded baited-hook methods so that smaller fish became relatively more intensively exploited. Although adjustments for these effects were made as well as possible by Herrington, quantitative comparisons between the period before and after 1930 appear uncertain. In particular, the relative level of recruitment indicated after 1930 seems too low to have sustained the continuing fairly high level of catch—that is, the 45-degree line indicated on Figure 24 probably does not apply to these later years.

What is of most interest for us is that the increase in rate of exploitation about 1930 was sufficient to eliminate oscillation from the population (Fig. 25). It was discovered in Figure 17 above that, in a stock subjected to increasingly heavy fishing, reproduction-curve oscillation may disappear while rate of exploitation is still rather low; and further, that the transition from marked oscillation to unrecognizably small oscillation occurs rather abruptly. Possibly accidental, but also in accord with expectation in a fishery of increasing intensity, is the fact that the second cycle in Figure 25 is shorter than the first (*two* years shorter in Herrington's representation).

Fruit Fly (*Drosophila melanogaster*)

An optimum population density for reproduction was found by R. Pearl and S. Parker in a series of experiments made during the early 1920's. The reproduction curve of Figure 26A is synthesized from graphs summarizing their work, given by Lotka (1925).

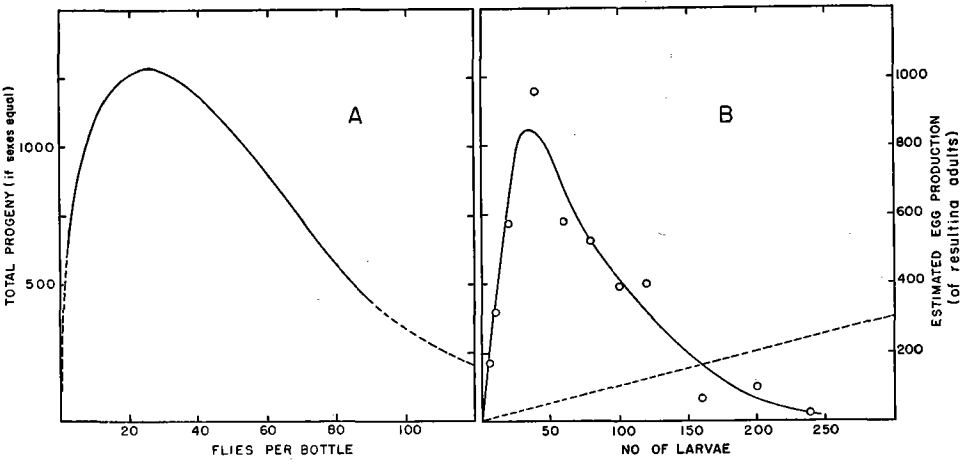


Figure 26. Reproduction curves for *Drosophila melanogaster*. A—From data of Pearl and Parker, as presented by Lotka (1925). The ordinate represents half the product of the density in question, times the progeny produced per female per day, times the mean length of life. The range over which observations were actually made is shown as a solid line. B—From data of Chiang and Hodson (1950, table 9 and fig. 13). The abscissa is the number of 24-hour larvae used to start a culture, and the ordinate is the potential egg deposition of the resulting adults. The dotted line is an approximation to the 45-degree line, which ignores possible mortality in the egg to 24-hour larva stages.

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Recently Chiang and Hodson (1950) have investigated *Drosophila* reproduction in more detail. Figure 26B shows their graph of "potential fecundity" against initial larval density. Some of the factors involved in decreasing the reproduction with increasing density were greater larval mortality, increased failure to pupate, and smaller size and lesser fecundity of adults produced. The flattening of the curve near the right end was due partly to a secondary *increase* in pupal and adult size at the highest densities.

Azuki Bean Weevil (*Callosobruchus chinensis*)

Extensive studies of this species have been made by S. Utida, but the only account I have been able to consult is that of Fujita and Utida (1953). There is no overlapping of generations. A reproduction curve is shown in Figure 27, obtained under experimental conditions. The right limb trends downward with a slope numerically less than unity, which would make the population subject to damped oscillations. Such oscillations were actually observed by Utida (Fujita and Utida, p. 494).

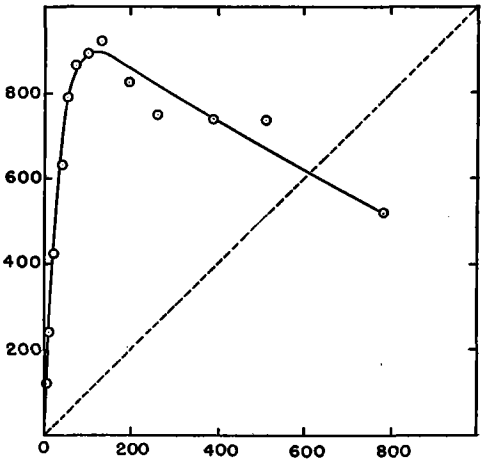


Figure 27. Reproduction curve for the azuki bean weevil under experimental conditions. *Abscissa*—size of parental generation; *ordinate*—number of mature progeny. Replotted from figure 4 of Fujita and Utida (1953).

Water-Flea (*Daphnia magna*)

Pratt (1943) performed experiments on the effect of stock density upon the production of young and upon longevity, in *Daphnia* grown in 50 cc. of culture medium with excess food. These can be used to construct reproduction curves, shown in Figure 28.

The ordinates in Figure 28 are equivalent to the total number of young produced in an average lifetime at the temperature and density in question (Pratt, p. 135). The identity of these values with those required for a reproduction curve can be shown as follows. Consider the life of a *Daphnia* to be divided into a series of "generations", corresponding to the years of an ordinary fish's life. These could be made of any length, but for convenience they are taken as equal to the approximate time from hatching to first maturity (6 days at 25°C., 14 days

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at 18°C.). Then consider a 10-*Daphnia* population at 25°. It produces young which live 26 days on the average (Pratt's fig. 6), or 4.3 "generations". During the first of these the *Daphnia* are immature, so that their mature life is 3.3 "generations". The total young produced by an individual in an average lifetime is, however, 23.1 (Pratt, p. 135), or 7.0 per generation. Consequently a population of 10 mature individuals produces $10 \times 7.0 = 70$ young per "generation", and these survive through 3.3 generations of maturity on the average, so that the total reproductive potential produced by the parent generation is $70 \times 3.3 = 231$ units. These "units" are numerically the same as the total number of young which would be produced throughout the life of the 10 parent individuals living at constant density. However the "units" are to be regarded not as young actually produced, but as a measure of the generation's potential contribution to the mature population of several future generations, in exactly the same way as eggs in mature females were used for fishes. In actuality, if stock density were to change by the time these 231 "units" were in the mature population, the actual production of young by the progeny of the original generation would be greater or less than 231.

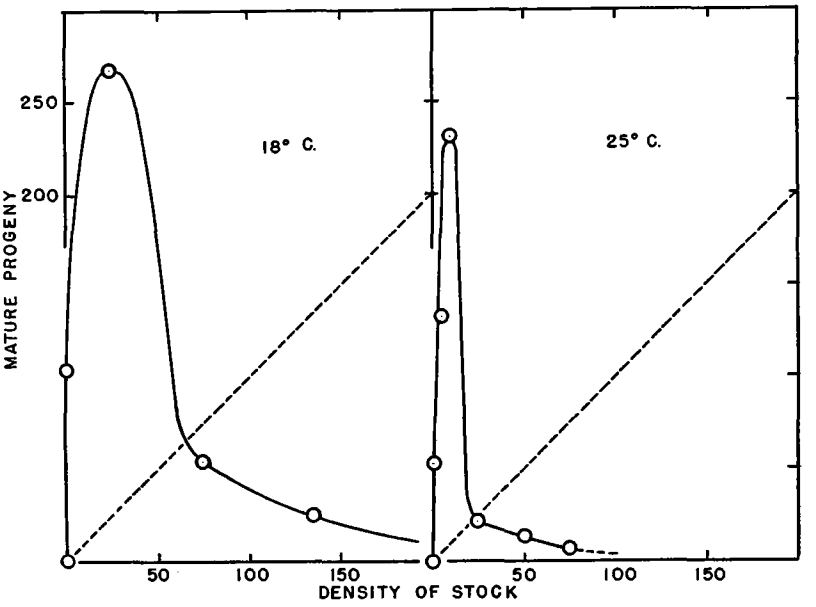


FIGURE 28. Relation between density of adult *Daphnia magna* and their expectation of progeny, at 18° and 25°C. Computed from data of Pratt (1943): 18° curve from his figs. 7 and 8; 25° curve from his figs. 5 and 7.

The reproduction curves of Figure 28 are of a steeper type than any of those discovered among fishes, particularly the 25° curve. Both curves are in fact likely to be even steeper than they are drawn, since the highest point of observed production of young is not necessarily the maximum. This steepness has been believed to be associated with the small volume of habitat and consequent likelihood of waste accumulation—although the medium was changed every second day.

Pratt (1943, figs. 1, 2) also followed the growth of *Daphnia* populations over a period of 6–7 months under the same culture conditions as outlined above. Starting in each case from 2 parthenogenetic females not more than a day old,

TABLE V. Simplified calculation of ideal population changes of *Daphnia magna* at 25°C. under the conditions of Pratt's experiments, by 6-day periods. Average mortality is considered as zero for the juvenile 6 days, one-fourth from the first to the second 6 days of maturity, one-third from the second to third, one-half from the third to fourth, and 100 per cent after the fourth. The number of "reproductive units" produced by a given stock in 6 days is defined as the product of the number of mature animals produced times the average number of 6-day periods each lives. Column 6 shows the total units produced by the stock of column 5, values being taken from Figure 29B. These units are divided among the appropriate periods to give the stock present, as shown in the first four columns. The population starts as two newly mature adults, one of which survives for two 6-day periods, the other for four. Total adult stock thereafter is the horizontal sum of the units in the first four columns. The total stock (column 7) is the adult stock (column 5) plus the number of juveniles, the latter being the same as the initial number of matures, as indicated in column 1 one 6-day period later.

1	2	3	4	5	6	7	1	2	3	4	5	6	7
Fraction mature during successive 6-day intervals							Fraction mature during successive 6-day intervals						
0.4	0.3	0.2	0.1	Adult stock	Reproductive units produced	Total stock	0.4	0.3	0.2	0.1	Adult stock	Reproductive units produced	Total stock
..	..	..	..	2	65	2	6	2	1	0	9	230	101
..	..	..	..	2	65	28	92	4	1	0	97	4	189
26	..	..	..	27	20	53	92	69	3	0	164	4	166
26	20	..	..	47	15	55	2	69	46	2	119	4	121
8	20	13	..	41	15	47	2	1	46	23	72	6	74
6	6	13	6	31	20	37	2	1	1	23	27	20	30
6	4	4	6	20	40	28	3	1	1	0	5	160	13
8	4	3	2	17	90	33	8	2	1	0	11	230	75
16	6	3	2	27	20	63	64	6	1	0	71	6	163
36	12	4	2	54	10	62	92	48	4	0	144	4	147
8	27	8	2	45	15	49	3	69	32	2	106	4	108
4	6	18	4	32	15	38	2	2	46	16	66	6	68
6	3	4	9	22	25	28	2	1	1	23	27	20	30
6	4	2	2	14	200	24	3	1	1	0	5	160	13
10	4	3	1	18	65	98	8	2	1	0	11	230	75
80	8	3	2	93	4	119	64	6	1	0	71	6	163
26	60	5	2	93	4	95	92	48	4	0	144	4	147
2	20	40	2	64	6	66	3	69	32	2	106	4	108
2	1	13	20	36	15	39	2	2	46	16	66	6	68
3	1	1	6	11	230	17	2	1	1	23	27	20	30

the 18° cultures rose to a peak of about 200 animals usually after about 70 days, decreased to about 120 animals, then showed a slight increase. At 25° the cultures grew more rapidly, reached a peak in 20 days, and then as rapidly declined; sometimes they died out, but in two instances the oscillation was repeated four times before the experiment terminated. Figure 29 shows one example at each temperature.

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The steep reproduction curves of Figure 28 immediately suggest the possibility of oscillations, and they have been used to obtain the computed population curves (B and D) of Figure 29. These start from 2 newborn animals, like Pratt's figures 1 and 2.

Construction of the curves of Figure 29 required certain assumptions and simplifications. The life of a *Daphnia* was divided into "generations" as before. The total number of "reproductive-units" indicated by the ordinate of Figure 28 was divided among successive generation-long periods of life as follows: for 25°C., in the ratio 0:4:3:2:1; for 18°C., in the ratio 0:22:20:18:14:11:9:5:1. These ratios correspond approximately to the average survival indicated by Pratt's figures 5 and 8. The calculation then proceeded on the same basis as in Table I, the sequence for 25° being shown in Table V. Because Pratt's censuses include immature individuals, the mature stock of column 5 is increased by the number of immatures in the culture (considered equal to the first generation of matures, in column 1 one line below) to give the total stock of column 7.

Obviously this computation differs from actuality in the assumption of an age distribution of mortality that is independent of density (cf. Pratt's figs. 5 and 8). The difference is most serious at 25°, and if a more complex calculation were made to take account of it, the initial effect would be to make the troughs deeper because of the shorter lives of denser stocks.

The predicted course of events in Figure 29 has considerable resemblance to what was actually observed in respect to period, and to a less extent in amplitude. (Pratt's observed curves also vary a lot among themselves, as would be expected when the stock starts out from, and at 25° is periodically reduced to, a very few individuals.) Any judgment concerning the significance of this resemblance would have to be of the subjective type which Smith (1952) rightly deplores, but the comparison has the merit that the factual bases of the two curves of each pair are from different sources—there are no "fitted constants".

At 18° the computed cycles are soon damped to an inconsiderable amplitude, but the suggestion of a second low peak in the observed series (Fig. 29A) would probably have been confirmed if the cultures had been maintained a month or so longer. At 25° the computed oscillation reached a stable equilibrium pattern having a large amplitude by the time of the fifth peak, and the two surviving observed series were still oscillating strongly when they were discontinued. The striking difference between 18° and 25°, in both the calculated and the observed curves, is ascribable partly to the steeper reproduction curve at 25°, partly to the fact that a larger number of "generations" are mixed in the 18° mature population.

However there is another factor which enters into these *Daphnia* fluctuations. At the higher temperature the young produced under crowded conditions are permanently less fertile than those from uncrowded cultures[10]. This phenomenon

[10]This was demonstrated by segregating newly mature specimens from crowded cultures, keeping them in isolation, and comparing their production with that of specimens reared singly, but the data cannot be applied quantitatively to these examples. Pratt did not perform the complementary experiment of crowding mature specimens which had been reared singly. However, the very sharp restriction of reproduction which occurs in such event is sufficiently indicated by the small number of progeny produced by the first young born in a culture. These live their adult (but not their juvenile) days under crowded conditions (cf. Pratt's fig. 3).

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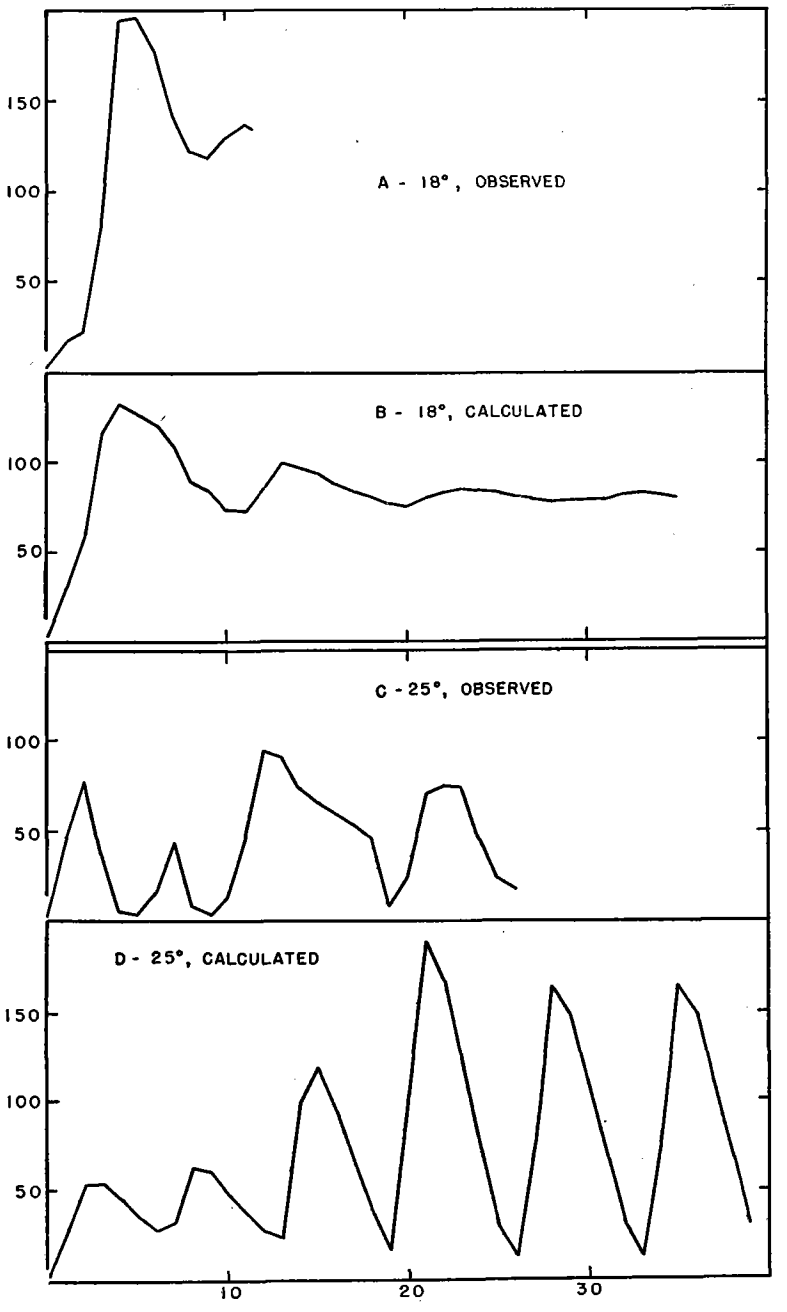


FIGURE 29. Observed and calculated fluctuations in *Daphnia* populations, from data of Pratt (1943). Curves A and C are directly from Pratt's observations, grouped by 14-day intervals at 18°C. and 6-day intervals at 25°C. Curves B and D are calculated from the reproduction curves of Figure 28. *Ordinate*—number of *Daphnia; abscissa*—successive 6-day or 14-day intervals.

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has been considered the sole basis for the oscillations observed at 25°, both by Pratt and by Hutchinson and Deevey (1949). The present analysis makes it seem more likely that this effect is not an essential feature of the system, and that *actual* density is the more important depressor of reproduction.

STARFISH (*Asterias forbesi*)

No reproduction curve is available for this starfish or any other. However Burkenroad (1946) describes a series of oscillations of its abundance along the New England coast, which have had an average period of 14 years over about 90 years. These seem sufficiently remarkable to justify some speculation concerning whether they could be ascribed to a steep reproduction curve like Figure 7 or 8. If so, the implication would be that the starfish make their median contribution to reproduction 7 years after the zygote stage, and also that they make a relatively small contribution during the first few years after they are hatched. During the peaks of its cycle *A. forbesi* apparently exists in the massive numbers which would make a steep outer limb seem plausible, having in mind that starfish readily eat their own young and that reproduction is partly suppressed among individuals of crowded and poorly fed populations (cf. Vevers, 1949).

The more doubtful aspects are whether starfish live as long as a 14-year cycle would imply, and whether there is sufficient time lag before reproduction. In respect to the latter, Galtsoff and Loosanoff (1939) found the most abundant size group of *A. forbesi* at Wood's Hole in April to have a mode at only 3 cm. (between tips of arms); these would scarcely mature the same year, and even the next group, with a mode at 7 cm., could not be very fecund. The maximum size for this species is about 20 cm., which suggests a life span of several years at least, since rate of linear growth presumably falls off rapidly at the larger sizes, particularly when the stock is dense. Concerning their year-to-year mortality rate little is known except that adult starfish seem to have few natural enemies, so that for the most part they should live out a normal physiological life-span, whose length would of course probably vary a great deal from individual to individual. While they live, the number of eggs produced by each presumably increases more or less in proportion to body volume, so the number produced by a given year-class probably reaches a maximum only after it has been in existence for several years.

On the whole it seems possible, if perhaps not too probable, that the median contribution of a year-class to reproduction could be made 7 years after it was hatched, which is what a 14-year reproduction-curve cycle implies. Because so many ages would contribute to spawning, the curve would have to be very steep —probably even steeper than Figure 8—in order to produce oscillations of the amplitude described by Burkenroad, i.e., having a peak to trough ratio of about 20 to 1. There would likely be minor complications resulting from differences in growth rate between times of abundance and times of scarcity, since growth of starfish is known to vary sharply in response to availability of food. Systematic sampling for size and (if possible) for age, over a period of one or more cycles,

should provide the key to the situation. Galtsoff and Loosanoff's samples were taken in 1936–37 at a low point in the cycle, and contained large numbers of young, as expected. At a peak of abundance large individuals should predominate, and young of the year should be very scarce toward the end of the year, or sooner.

Sinuous Reproduction Curves

Dr. Pritchard's data on pink salmon reproduction at one site have already been summarized, as well as the adult counts made on the Karluk River. More recently, Neave's (1952, 1953) review of pink salmon reproduction in a number of localities has led him to suggest a rather complex reproduction situation. The species is peculiar in that there are a few areas where one of the two "lines" is completely absent in a series of streams. In other areas one line exists at a level much below the other, and the two lines will have maintained approximately the same relative position for decades, though not necessarily for the whole period of record. In still other areas there is no consistent difference between the lines in respect to abundance. Furthermore there have been cases where one line has dropped suddenly from a high to a low level of abundance and has remained at that level for a considerable period; in one or two instances such a reduced stock has bounced back up to its former level.

Dr. Neave feels that density-dependent mortality in this species must occur mainly during its freshwater life—from the time the adults appear in spawning streams in August-October to the time the fry leave in April or May. Further, there are factors causing both compensatory and "depensatory" mortality at this stage, the latter being mortality which kills a relatively larger fraction of the fish present when their numbers are small than when they are large. Experiments have shown that the activity of stream predators (trout, etc.) upon pink fry is depensatory —the fry apparently being so vulnerable that the predators eat all they can, then leave the rest alone.

This hypothesis accounting for two fairly discrete and fairly stable levels of population in pink salmon implies the existence of anomalous reproduction curves; two possible types are shown in Figures 30 and 31. (The part to the right of the dome does not enter into the argument.) In Figure 31, at some intermediate level of stock, depensatory mortality is at a maximum relative to compensatory mortality, producing the dip in the left limb of the reproduction curve. For example, a population which adhered closely to the curve of Figure 30 could be in equilibrium with a fishery taking 50 per cent of the adult fish at the two different levels of abundance indicated, but between those levels equilibrium rate of exploitation falls to as low as 20 per cent. Since 50 per cent is the maximum permissible limit of exploitation at the higher level of stock, any slightly greater exploitation will send the stock tumbling to the lower level. There it will stay unless exploitation is sharply restricted or temporarily discontinued—merely reducing it to 50 per cent again will not restore the higher equilibrium position.

More generally, when random variation in reproductive success is taken into account, the existence of a fishery makes it easier for a population to slip down

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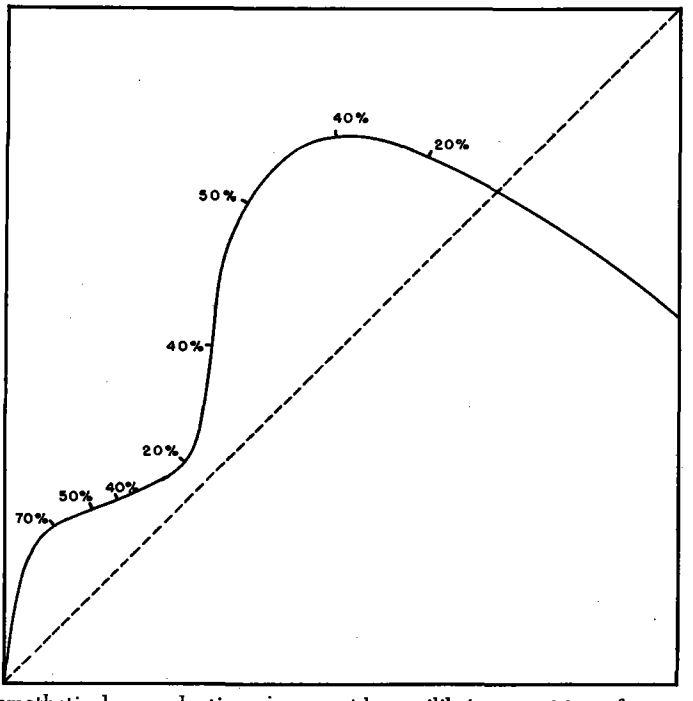


FIGURE 30. Hypothetical reproduction curve, with equilibrium positions for several rates of exploitation. See text.

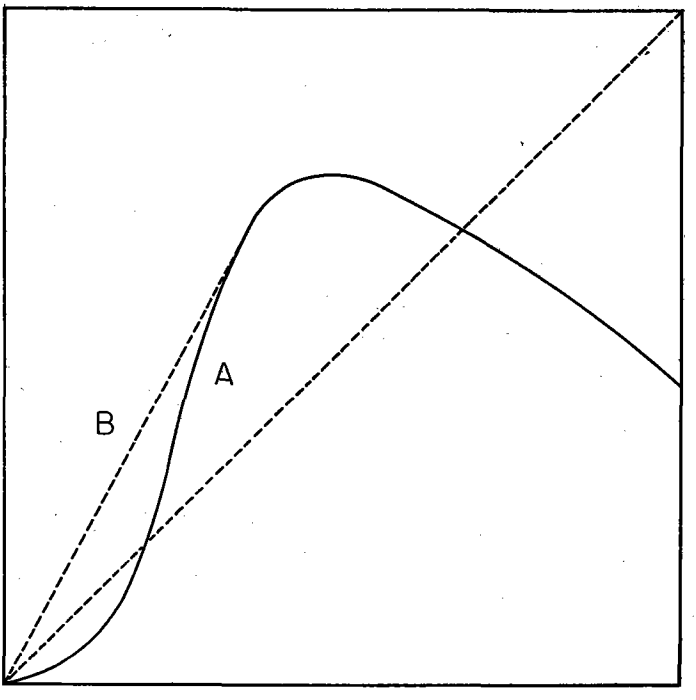


FIGURE 31. Hypothetical reproduction curves. See text.

from the higher to the lower abundance level in a bad year, and harder for it to get back up in a good one. Of course, if the reproduction curve falls off to the right of the 45-degree line, as is actually suggested in Figure 30, the *first* effect of a new fishery would be to increase population abundance.

Figure 31A differs from 30 in that there is no lower level of strong resilience to exploitation. It might represent pink salmon reproduction in areas where one of the two lines ("cycles") does not exist at all. Accidental reduction of the population past a certain low level in that case initiates a progressive decrease and results in eventual extinction. A similar situation is implied in a hypothetical net-change curve of Haldane (1953, fig. 3), and for several animals there is some experimental evidence of reduced effectiveness of reproduction at very low densities (cf. Allee, 1931; Hutchinson and Deevey, 1949).

Dr. Neave's ingenious hypotheses, illustrated by the above curves, are supported by several lines of evidence, but have not as yet been firmly established for any pink salmon stock. McClinton Creek, described earlier, is a stream which has no pinks at all in odd-numbered years, hence might be expected to have a reproduction curve like Figure 31A. The actual curve, however, is of a simple type as far as it is known (Fig. 19). If the left-hand portion of this curve falls below the 45-degree line, it must do so at densities considerably less than 5,000 female spawners—that is, in the extreme lower left corner of Figure 19B. At Karluk there is some indication of an inflexion of the curve at low stock densities (Fig. 20, upper right), but it is not clear that it is sufficient to play the role described above.

A difficulty with the curves of Figures 30 and 31A is that their initial phases are quite different, and hence it seems improbable that both should be applicable to the same species in the same general region. They might be reconciled if the beginning phase of 31A, in varying degrees of development, were commonly found tacked onto the initial part of the Figure 30 curve, producing a doubly sinuous ascending limb for the curve as a whole. If so, hypotheses as to the nature of the compensatory and depensatory mortality would have to be correspondingly more complex. A simpler but less satisfactory compromise type would have merely a straight left limb, as in Figure 31B; this would account for any observed lack of resilience in a population after a certain level of exploitation had been reached, since one density would be as stable as another once the equilibrium point had been shifted to the ascending limb of the curve.

Discussion

Doubtless other investigators will be able to add to the stock of reproduction curves from data now available, and we may expect more in the future. Considering the curves for fishes, cited above, it is unfortunate that in most cases only a section of the curve is available.

A steep ascending limb, as in Figure 2b or 8, is implied whenever a fishery consistently takes some large fraction of a stock before it has a chance to reproduce, as notably among various salmon stocks. Actual points on this ascending

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limb are available in a few instances where a population has risen from or declined to a very low ebb; the Shuswap sockeye are the best example.

No curve yet discovered suggests that the maximum of recruitment is at or to the right of the 45-degree line (as in Figs. 1, 3, 4), but on several of the curves this line cannot be located.

Concerning the right limb, an indefinite horizontal extension like that of Figure 2 has often been suggested for sea fishes on *a priori* grounds (e.g., Baranov, 1918; Kesteven, 1947). Tester (1948) found this hypothesis consistent with the Vancouver Island west coast herring data, and this is still true (Fig. 18). The same can be said of some of the other curves examined (Figs. 19B, 20, 22). However, without any exception, the lines which best fit these outer limbs slant downward at least slightly, and the weight of their combined evidence

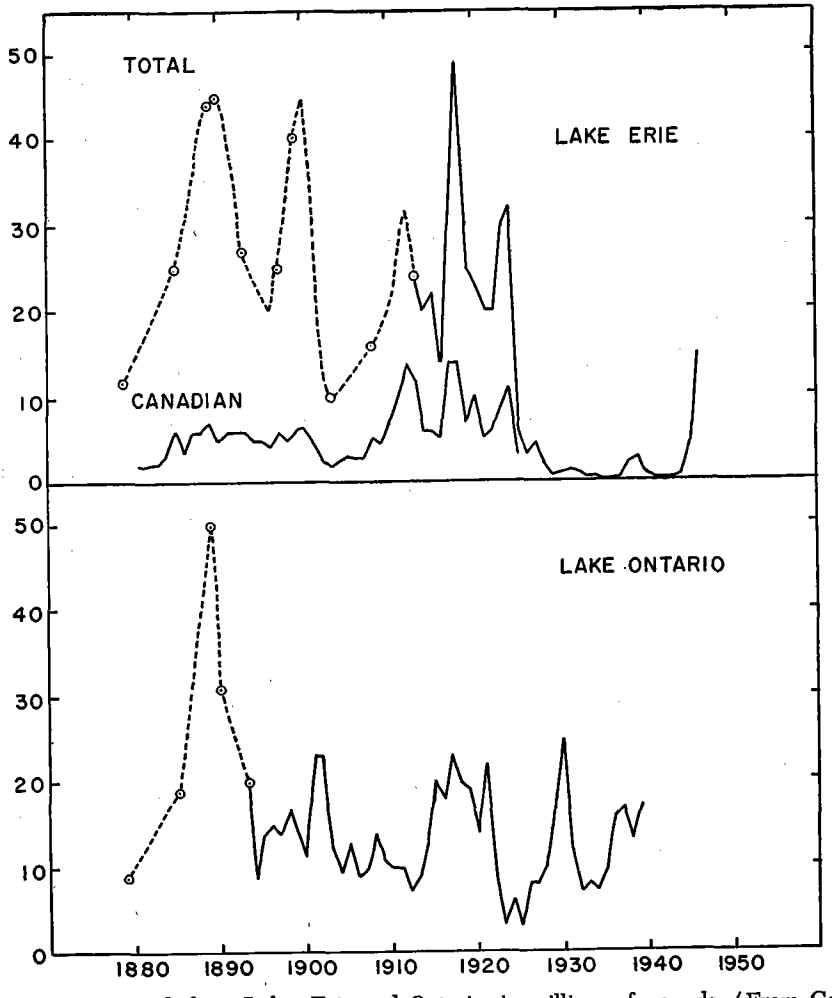


FIGURE 32. Cisco catch from Lakes Erie and Ontario, in millions of pounds. (From Gallagher *et al.*, 1943; and Scott, 1951.)

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indicates that negative slopes of at least moderate magnitude are real and are the common situation. On the other hand, an approximation to the flat curve of Figure 2 might well be expected in a species like the coho, which is aggressively territorial and whose stream habitat is limited.

The best-documented example of a really steep descending limb in a natural fish population is provided by the 1912–29 New England haddock (Fig. 24), where the curve is approximately like Figure 7. Even steeper curves characterize the invertebrate populations (Figs. 26–28).

Left to themselves, steep reproduction curves produce such beautifully regular oscillations in population abundance that it is rather disappointing to discover how easily this regularity is disturbed by non-compensatory mortality (Fig. 11B). This means that we should rarely expect to be able to identify the contribution of a reproduction curve to any given series of observations of animal abundance, even though such contribution may be a major factor in many examples of marked fluctuation. And it is a fact that most of the well-known examples of fluctuating abundance cannot be a *simple* resultant of a steep reproduction curve. The 9- or 10-year cycle of grouse and hares, for example, could not be of the simple type shown in Figure 10 because there is not the necessary lag in reproduction—the young are mature in the next reproductive period after they are born; in any event it is doubtful if average longevity in these species is sufficient to put a brood's mean time of contribution to reproduction at 4.5 or 5 years. Certain major changes in fish populations—for example, cod off Greenland, herring off Norway, or mackerel off eastern North America—have occurred on too great a scale and with too long a period to have been governed by any single reproduction curve. Fish population changes having shorter period, like those of the ciscoes of Lake Ontario (Fig. 32), might be strongly influenced by their reproduction curve, but the peaks and troughs are too irregular, and basic population information is insufficient, to permit even a good guess. If the cisco rate of exploitation is moderately great, as seems probable, then direct reproduction-curve periodicity is not likely to be recognizable (cf. Fig. 17).

We are perhaps lucky to have, in Herrington's haddock, even one example of an oscillation in which the effect of the reproduction curve can be identified with fair likelihood.

OTHER REPRODUCTION SITUATIONS

Up to this point all examples and interpretations have been predicated on the postulates that it is only animals of mature size whose abundance affects compensatory mortality; and that where there is human exploitation of the stock, it takes only mature individuals. What is the effect of relaxing each of these restrictions?

Larger Immature Fish Taken by the Fishery

If a fishery takes fish of smaller than mature sizes, it makes little difference to the theory of population control outlined above *as long as the sizes subject to compensatory mortality are not touched.* Details of the equilibrium level of

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population at different intensities of exploitation are of course affected, but not the general course of events. In actuality it has fairly often been found that spawning populations can be reduced to a very low level while the catch is maintained at a fairly high level by immature fish, as for example the cod in the North Sea. In this and similar instances data for plotting a reproduction curve do not seem to be available, but the general situation strongly suggests that the right limb of the curve must slope downward and that the left limb must be very steep. Until reproduction curves are available and future recruitment can be predicted, proposals for increasing catch by reducing the fishing intensity in the North Sea and elsewhere will lack that final symmetry which would make them wholly convincing.

Exploitation during the Compensatory Phase

Exploitation that takes fish at an age when natural mortality is still compensatory means, for practical purposes, a fishery for young during the first year or two of their life—the earlier the better. The removal of such young is at least partly balanced by increased survival and/or growth of the remainder; in fact, the effects of removals at this stage are equivalent to reduction of the spawning stock which produced the brood in question. If the reproduction curve for the population is of any of the types 3–8, such reduction will at first *increase* net production of recruits, which will produce more eggs and permit a larger catch of young in future years. This ascending spiral of abundance may continue until the level of stock is reached which produces maximum recruits.

There can of course be no general rule indicating a single optimum time for exploitation for all fisheries. In any particular instance, maximum yield may be obtained by taking young, or adults, or both; and in balancing the alternatives, consideration should be given to the relative value per pound of the two sizes and the ease with which they are caught. But it is clear that any *general* prejudice against exploiting young fish is unsound. Each case should be considered on its merits. The situation most favourable for juvenile exploitation is evidently that where the fish reach a *relatively* large size before compensatory mortality ceases to be important, so that the catch taken during this period can be large in total bulk.

Relatively few fisheries now exist in the temperate parts of the world which attack really young individuals, but this can be ascribed mainly to the (usually) lesser value per pound of small fish as compared with large ones, and the (usually) greater difficulty of catching these young. In Japan a number of species are eaten as larvae—for example the sand lance (*Ammodytes*). There and elsewhere the exploitation of various clupeids in post-larval, but still early, life is fairly common.

One such fishery on this continent is that for "Quoddy" herring in southern New Brunswick (Huntsman, 1952, 1953). A large primitive stock of mature herring in this region apparently fluctuated in abundance during the middle years of the last century, but no details of period or amplitude are available. About 1880 mature herring declined rather abruptly and they have not reappeared in

comparable numbers since. About the same time the weir fishery for "sardines" (age 0 to age II herring, mainly) developed enormously and has yielded large and fairly steady catches ever since—catches which bulk two or three times as great as did the early fishery for mature individuals.

If we postulate that the weirs take the young partly during the time that compensatory mortality is operative, and partly after compensation is no longer important, a formal explanation of this situation can be made. The sardines taken young, while the mechanism of compensation is still in operation, represent a diversion to human use of stock which would otherwise be lost to predators or other natural causes. The sardines taken at larger sizes, after compensation is largely finished, plus any larger stock taken, reduce the potential breeding population to a point far below the numbers it would otherwise attain (and did actually attain in years gone by). This reduced spawning stock produces many more young fish than a large one would, the implication being that in decreasing in abundance it has climbed the right limb of a dome-shaped reproduction curve such as Figure 8. In this way the present distribution of sizes and numbers could be maintained.

Compensation by Immature Members of the Population

In the restricted habitat of ponds and small lakes cannibalism is one of the most likely methods of population regulation for abundant predacious species, e.g., bass (*Micropterus*) or crappies (*Pomoxis*). Indeed, the first stages of all centrarchids are so tiny that even a normally insect-eating species like the bluegill (*Lepomis macrochirus*) may eat eggs or fry of its own species during their very early life. Immature individuals tend to be most active in this, since they normally frequent the shallow water where most spawning occurs, and gather around nests of their own or other species. Adults are less frequently in contact with young, and the males actually guard eggs and young as long as they are in the nest, or even longer; hence adults are less effective in reducing the oncoming generation. As an example, two similar ponds at the Tri-lakes Hatchery, Indiana, were stocked in spring with similar weights (13–15 kg.) of bluegills: in Pond 2 they consisted almost wholly of mature fish, in Pond 4 about half (by weight) were small fish of the previous year's brood. By November the young of the year produced by Pond 2 numbered 19,150 and weighed 36.5 kg., whereas Pond 4 produced only 230 fingerlings weighing 0.18 kg.

To adequately illustrate situations of this sort it would be necessary to construct a reproduction diagram showing the three-way relationship between quantity of mature stock, quantity of immature stock, and resulting recruitment. Present information does not permit this, but the general effect of participation of immatures in compensation must be to spread the reproduction-depressing influence of any successful brood over a longer period of time, and to make it almost impossible to have the "lag" effect which produces population oscillation. In such a situation, however, cycles of fish *size*, and to some extent of poundage, might occur if individuals of a successful brood nearly all died off during the same calendar year (cf. Thompson, 1941, p. 209).

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TOWARD A THEORY OF RECRUITMENT

The justification for using any reproduction curve must in the long run come from observation. However it would be most useful, if it were possible, to formulate some general theory of reproduction which might lead to a standard type of reproduction curve applicable in a majority of situations.

THEORY OF PREDATION

An approach to this goal can be made by way of a consideration of predation upon the young of a species, whether by other animals or by older individuals of its own species. The theory of predation was briefly considered by the writer in a recent paper (1952), and the following quotation will provide a basis for the further argument here:

> It is convenient to distinguish three types of numerical relationship between predators and a species of prey which they attack.
>
> A. Predators of any given abundance take a fixed number of the prey species during the time they are in contact, enough to satiate them. The surplus prey escapes.
>
> B. Predators at any given abundance take a fixed fraction of prey species present, as though there were captures at random encounters.
>
> C. Predators take all the individuals of the prey species that are present, in excess of a certain minimum number. This minimum may be determined in different ways: (1) There may be only a limited number of secure habitable places in the environment, so that some prey are forced to live in exposed situations where capture is inevitable. The number of such secure niches may be partly governed by territorial behaviour of the prey. (2) The maximum "safe" density of prey may be the one at which predators no longer find it sufficiently rewarding to forage for them, and move to other feeding grounds.
>
> The three situations above tend to intergrade, of course, but it is useful to keep their differences in mind.

SITUATION A

This is likely to occur when a prey species is temporarily massed in unusual numbers, for example, adult herring in spawning schools, or newly-emerged fry of pink and chum salmon going downstream. The main characteristic of such situations is that the number of prey eaten depends on the abundance of predators, but not on the abundance of prey. Hence such situations cannot last long, and the predators cannot make the prey in question their principal yearly food; otherwise they would almost surely increase in abundance and the situation would change to type B or type C. If a type A situation persisted for long, it would come to an abrupt end with the extermination of the prey.

SITUATION B

Here the number of the prey species eaten is proportional to the abundance of predators and to the abundance of prey. Unlike A, this type of predation can easily occur over long portions of the year; and the prey species may comprise the larger portion of the predators' annual ration. This situation was observed at Cultus Lake, British Columbia, for predation of squawfish, char, coho and trout upon fingerling sockeye, over a wide range of abundance of the latter (Foerster and Ricker, 1941).

SITUATION C

The classical example of this situation was described by Errington for bob-white in Iowa, where a given range would winter safely a fixed number of birds, practically independently of the number which were present in autumn, the surplus being taken by predators. Studies conducted from the Atlantic Biological Station of the Fisheries Research Board of Canada,

St. Andrews, N.B., suggest that in some rivers predation upon Atlantic salmon parr tends toward type C, because a marked increase in the number of young fish planted was followed by only a relatively small increase in number of surviving smolts (Elson, 1950). In type C situtions the predators must tend to have ample alternative foods, and they are often mobile so that they can leave an area where the food supply has been "cleaned up" for the season.

Consider a prey species which is subject to type B predation over some part of its life history. Assume first that predation causes the whole of the mortality that the prey is subject to during that period. Then the prey species decreases in abundance according to the well-known exponential formula

$$N/N_0 = e^{-it}, \tag{1}$$

where: N_0 is initial abundance, N is abundance at time t, $e = 2.718 \ldots$, and i is a statistic representing the fraction of the prey which would be eaten in a unit of time if its abundance were held constant for that long; i is often called the *instantaneous* mortality rate.

Under the conditions postulated, instantaneous mortality rate is directly proportional to the abundance of predators. This can be illustrated by considering a situation where predators attack a prey population of 1,000,000 individuals under type B conditions, and they inflict losses corresponding to $i = 0.8$, where the unit of time, t, is the whole season that predator and prey are in contact. Equation (1) indicates that in, for example, 1/1000 of the season, the predators will eat $i/1000 = 0.0008$ of the prey present, or 800 fish. During the next thousandth of the season, the predators eat $i/1000$ of the surviving prey, or $999{,}200 \times i/1000 = 799$ fish. The following interval they eat $998{,}401 \times i/1000 = 799$; then $997{,}602 \times i/1000 = 798$. This continues until all the thousand time-intervals have elapsed, at the end of which there are:

$$1{,}000{,}000(1 - 0.0008)^{1000} = 472{,}400$$

survivors, or 47.2 per cent. What happens if the number of predators, is doubled? In that event, during the first thousandth of the season twice as many fish will be eaten, i.e., 1,600, leaving 998,400. In the next thousandth the fraction eaten is likewise double, namely 0.0016; multiplied by the number of survivors this gives 1,597; and so on. At the end of the season $1{,}000{,}000(1 - 0.0016)^{1000}$ survive, which is 201,900, or 20.2 per cent.

Thus doubling the number of predators doubles the instantaneous mortality rate (which is true generally), but it increases actual mortality by only 27.0/52.8 = 51 per cent (which is true of only this particular example).

The general relation between predator abundance and actual mortality, or survival, is most conveniently expressed as:

$$\frac{p_2}{p_1} = \frac{\log s_2}{\log s_1}, \tag{2}$$

where p_2 and p_1 represent two levels of predator abundance, and s_2 and s_1 are the corresponding survival rates for the prey. This expression can easily be derived from (1), since p is proportional to i, and $s = N/N_0$ when $t = 1$.

Cannibalism

Of all the methods of population regulation listed earlier, cannibalism is the one in which the abundance of the control agent is most closely and inseparably allied to that of the population controlled. That is, an increase in mature stock not only increases the number of eggs laid or young born in a given reproductive season, but it also decreases the rate of survival of those young. What is the combined effect of these two opposed influences?

Let the size of a stock be measured by the number of eggs it lays, and let this size be proportional to its *instantaneous* efficiency in cannibalism (the fraction of young eaten by the adults in a short interval of time). Consider two situations characterized by different sizes of stock and let the ratio of the second to the

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first be w, corresponding to the p_2/p_1 of equation (2) of the quotation above. Finally, let all other sources of mortality, whether they occur before, after or during the period of the cannibalism, be density-independent, their total effect being to reduce survival rate of eggs and young to the fraction k of what it would otherwise be. The number of eggs laid, in situation 1, is E_1; in situation 2 it is therefore $E_2 = wE_1$. The reproduction (absolute number of recruits produced) in situation 1 is:

$$R_1 = E_1 ks_1; \tag{3}$$

and in situation 2 it is:

$$R_2 = E_2 ks_2 = wE_1 ks_2. \tag{4}$$

From equation (2) of the quotation above,

$$w = \frac{\log s_2}{\log s_1}; \quad \text{hence } s_2 = s_1{}^w. \tag{5}$$

From (4) and (5), the reproduction in situation (2) becomes:

$$R_2 = wE_1 ks_1{}^w. \tag{6}$$

Expressed as a fraction of the reproduction in situation 1, this is:

$$ws_1{}^{w-1}. \tag{7}$$

Plotting numerical values of (7) against w, for various values of s_1, yields a family of curves each of which has an origin at zero, a dome, and an extended right limb which approaches the abscissa asymptotically. No matter what survival rate is chosen for s_1, each curve describes the whole range of possible survival values as w is varied; hence only one curve of the family is needed. The convenient one to choose is that for which (7) is a maximum when $w = 1$. To locate this maximum, (7) is differentiated with respect to w and equated to zero:

$$ws_1{}^{w-1} \log_e s_1 + s_1{}^{w-1} = 0; \tag{8}$$

whence,

$$-\log_e s_1 = 1/w; \quad \text{or, } s_1 = e^{-1/w} \tag{9}$$

The value of s which makes (7) a maximum is $e^{-1/w}$, and if this is taken as the initial value of s (that for which $w = 1$), s becomes equal to $1/e$ or 0.3679. Substituting $1/e$ for s_1 in (7), we thus finally obtain the expression

$$we^{1-w}. \tag{10}$$

This shows the actual level of reproduction as a fraction of the maximum, when w represents the ratio of the actual density of mature stock to the density which gives maximum reproduction. Values of (10) are plotted as curve B of Figure 33.

From the equations and the curve the following conclusions are evident:

1. Since the curve approximates to the abscissa as w is increased, then *if the breeding stock is made sufficiently large, cannibalism reduces reproduction practically to zero*, in spite of the greatly augmented egg deposition. In theory this is true no matter how small the original rate of cannibalism; in practice, a type B predation situation between adults and young could not be maintained over *too* wide a range of stock densities.

2. Since, as w approaches 0, e^{1-w} approaches e, it follows that at minimal densities the survival rate from cannibalism is e times the survival characteristic

of maximum reproduction. In other words, the instantaneous mortality rate at maximum reproduction is greater by unity than it is at vanishingly small stock densities (since $-\log_e (1/e) = 1$; cf. expression (1) of the quotation above).

Curve B of Figure 33 can be changed into a reproduction curve by changing the scale of each axis to represent actual numbers of parents and progeny measured in comparable units—for example the egg production unit described in an earlier section. Equation (10) gives no information concerning the steepness of this actual reproduction curve, which will depend chiefly on the magnitude of k, the survival rate from factors other than cannibalism. On Figure 33 a number of possible curves have been drawn, all based on equation (10) but using different ratios of abscissal to ordinate scales. Referring each to the 45-degree line indicated, shapes reminiscent of several of the arbitrarily drawn types of Figures 3–8 can be identified, as well as most of the curves observed in actual populations. Curve E, of course, could not describe a stable population because it lies well below the 45-degree line, while even D would be rather precarious.

In addition to cannibalism, the argument of this section can be extended to the analogous situation where strife between mature and young animals occurs and results in frequent killing of the latter, but without necessarily any eating of one by the other; this has been described among muskrats, for example (Errington, 1954). In so far as these contacts are of type B—with the older animals killing a fixed fraction of the young they encounter—they would have the same population status as described above.

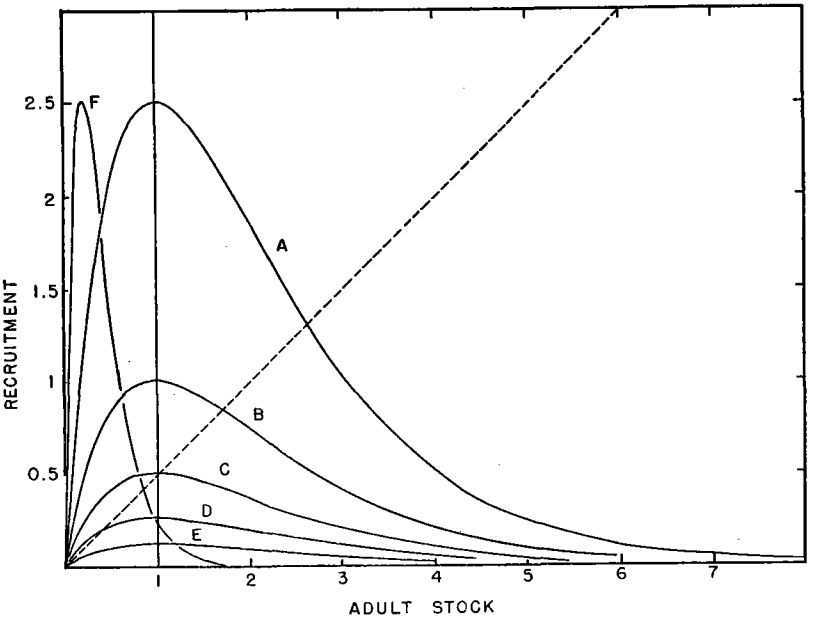


FIGURE 33. Graphs of we^{1-w} ("recruitment") plotted against w ("adult stock"). Curve B corresponds to the axes as labelled; the other curves are obtained by varying the ordinate and (for F) abscissal scales.

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It is not easy to assess the probable importance, in nature, of cannibalism and allied effects. In most of the fish populations whose reproduction curves were examined earlier there is little or no evidence of cannibalism or of other direct injury to young by their parents, and among *Oncorhynchus* this is impossible because the parents die. However, in assessing the possible role of cannibalism (or any other agent of predation) in population regulation, it is not possible to weight it directly on the basis of the number of units which it kills. The latter will depend mainly upon the stage at which it is active. For example, suppose that cannibalism by mature trout fell upon their fingerling progeny following an average density-independent egg mortality of 50 per cent and fry mortality of 90 per cent (95 per cent in all); and that the adult trout ate 63 per cent of the 5 per cent remaining, or 3 per cent of the original brood. In that event cannibalism would seem to be quite insignificant either from the point of view of the percentage of the total young which it killed, or from the point of view of the frequency of occurrence of small trout in stomachs of larger trout. In spite of this, under the conditions just postulated cannibalism would be the sole mechanism regulating the abundance of the population.

Predation by Other Organisms

Even granting that lethal contact between adults and young is the most direct and least fallible population-regulating mechanism, and that it is quite apt to exist undetected, the writer's present opinion is that it will probably not prove to be important in more than a minority of populations. In its absence, other compensatory agents must take over the role of regulators which determine the shape of the reproduction curve.

TYPE B SITUATIONS. A predator can qualify as an agent of population control only if it can increase in density or effectiveness as the abundance of prey increases, and vice versa. One way this can occur is by migration of additional predator units to the region of predation when prey abundance is large. For example, greater-than-average abundance of herring spawn might attract birds from a wider area than usual. It is possible (not necessarily probable) that, after gathering together, such aggregations would remain in the vicinity for a few days longer than the supply of eggs justified it, with the result that an initially superior spawning would yield a less-than-average number of eggs hatched.

A predator might also actually increase its total abundance with that of the prey in question. Consider, for example, a plankton predator or assemblage of predators which feeds on the pelagic eggs and young of mackerel. Suppose that in years when mackerel eggs approach zero abundance these predators (which have additional foods) are numerous enough to consume the fraction $(1 - s_1)$ of the eggs. Suppose also that for each unit increase in egg numbers the average predator population increases by one-tenth of its "basic" number. (This average would be taken over the whole time that predators and mackerel are in contact because, just before the eggs are available, initial predator abundance is assumed to be the same in all years.) During the later stages of the predator-prey contact the abundance of predators would have "overshot" that of the prey, with the

result again that mackerel broods initially larger than average would end up smaller than average. From expression (2) or (7) it is easy to calculate that relative reproduction is equal to

$$Zs_1^{1+Z/10}, \qquad (11)$$

where Z is the density of mackerel eggs immediately after spawning in a given year, in the unit mentioned above, and the predator density corresponding to no eggs is taken as unity. Values of (11) are calculated in Table VI, which shows the relative number of mackerel surviving the pelagic stage, for a series of initial prey densities and for two values of s_1. Sette (1943) estimated total survival of

TABLE VI. Survival rate and absolute number of survivors when average predator abundance varies with initial prey density in the manner indicated by columns 1 and 2. Pairs of survival rates and survivor numbers are shown for initial survival rates of 0.1 and 0.0001. The last two rows show the computation of the maximum number of survivors for each situation, which occurs when survival is 0.3679 ($1/e$) of its initial value.

1	2	3	4	5	6
Prey abundance	Predator abundance	Survival rate of prey	Relative number of survivors	Survival rate of prey	Relative number of survivors
			(1) × (3) × 10^3		(1) × (5) × 10^7
0	1.00	0.1000	0	0.0001000	0
0.2	1.02	0.0955	19	0.0000832	166
0.4	1.04	0.0912	37	0.0000692	277
0.6	1.06	0.0871	52	0.0000575	345
0.8	1.08	0.0832	67	0.0000479	383
1.0	1.10	0.0794	79	0.0000398	398
1.2	1.12	0.0759	91	0.0000331	397
1.4	1.14	0.0724	101	0.0000275	385
1.6	1.16	0.0692	111	0.0000229	367
1.8	1.18	0.0661	119	0.0000191	343
2.0	1.20	0.0631	126	0.0000158	317
2.5	1.25	0.0562	141	0.00001000	250
3.0	1.30	0.0501	150	0.00000631	189
4	1.4	0.0398	159	0.00000251	100
5	1.5	0.0316	158	0.000001000	50
6	1.6	0.0251	151	0.000000398	24
8	1.8	0.0158	127	0.000000063	5
10	2.0	0.0100	100	0.000000010	1
15	2.5	0.0032	47	...	...
20	3.0	0.0010	20	...	...
30	4.0	0.0001	3	...	...
1.086	1.1086	...	...	0.00003679	399
4.343	1.4343	0.03679	160	...	...

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Atlantic mackerel eggs and pelagic larvae, in 1932, to lie between 0.000001 and 0.00001, so that the $s_1 = 0.0001$ column of Table VI would not be unrealistic.

The distributions of columns 4 and 6 of Table VI are of course the same as those of Figure 33. Expression (11), like (7), is a maximum when the rate of survival from the predator in question is $1/e$ of what obtains when egg abundance is close to zero (cf. Table VI). Thus we again conclude that *the instantaneous mortality rate at maximum reproduction is greater by unity than the rate characteristic of a very small population density.* For example, if $s_1 = 0.01$, the instantaneous mortality rate at minimum stock density, from the action of the controlling predator, is equal to $-\log_e 0.01 = 4.61$. At the density of maximum reproduction, mortality from this predation becomes 5.61. Mortality from other factors ($= -\log_e k$) must be added to this 5.61 to give the total average egg-to-egg instantaneous mortality rate at maximum reproduction. Note however that at the *replacement* density of stock the compensatory mortality is greater than 1, its exact value depending on the steepness of the reproduction curve (relative to the 45-degree line).

The form of expression (11) indicates that the magnitude of the arbitrary factor 10, relating initial egg density to mean predator density, does not affect the general shape of the recruitment curve, though it of course affects its steepness: the larger this factor, the broader is the range of initial egg densities which afford substantial reproduction. Furthermore the rule italicized above holds even if the relation between prey and predator is quite irregular.

Similarly, the magnitude of s_1 affects the shape of the reproduction curve only in that the smaller s_1 is, the greater is the percentage change in reproduction produced by a given change in predator abundance (Table VI).

SITUATIONS OF TYPES A AND C. So far the theory presupposes type B predation situations. Type A situations need not be discussed in detail: for reasons given earlier, they tend to be restricted in space or time, and they lack the qualifications of a population control mechanism. Their effect on reproduction curves would be to introduce irregularities such as are shown in Figures 30 and 31.

Type C situations however *can* regulate a population, and in a pure form they produce reproduction curves like Figure 2b. The horizontal part of the curve corresponds to the limit of surviving young which the habitat will sustain. Type C situations may grade into type B, in which case it would in practice be difficult to distinguish a broad-domed type B curve from the mixed type.

It is also possible for type B and C situations to follow each other, and this is examined in Figure 34. If the type B situation comes first, its typical reproduction curve is truncated by the limit of reproduction imposed later by Type C (curve A of Fig. 34). If the type C situation comes first, it will at a certain point limit the brood to a density which may be less than, equal to, or more than what gives maximum reproduction in the subsequent type B situation; these three possibilities are illustrated by curves B–D of Figure 34. Note that while the type C situation can *reduce* the difference in instantaneous mortality rate between minimal and maximal reproduction, it cannot *increase* this difference.

The various curves of Figure 34 should be looked for in nature when the biology of the animal in question points in their direction. Among the examples given earlier, a limit of environmental capacity is strongly suspected for coho salmon; actually Figure 21 could readily be fitted with a truncated curve like 34A, but there is only one point for the outer decreasing phase.

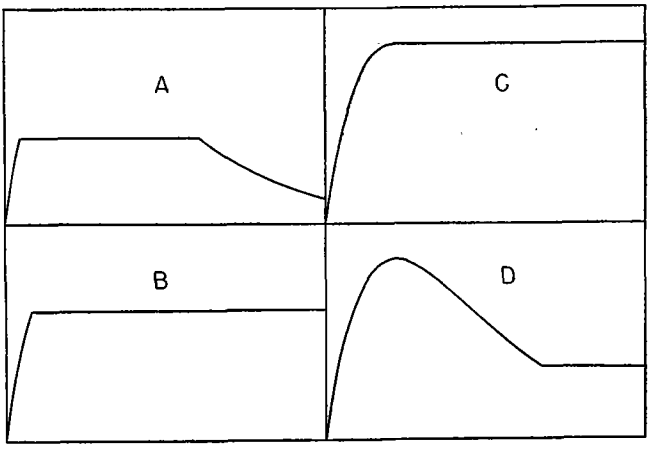


FIGURE 34. Reproduction curves when predation situations of types B and C occur in succession. *Curve A*—when type B precedes type C; *curves B–D*—when type C precedes type B. The type B reproduction curve is the same for all four situations.

SIZE OF PREDATOR AND SIZE OF PREY. It has sometimes been said that large predators (e.g., foxes) cannot control or regulate the abundance of a smaller prey species (e.g., mice), because the rate of reproduction of the prey is so much greater than that of the predator. This argument would be valid only if the predation situation were of type A, with the foxes always able to kill as many mice as interested them—a situation which could scarcely last indefinitely. However if the fox-mouse predation situation were of type B, calculations from (7) show that any increase at all in the number of foxes reduces the survival rate of mice more than proportionately, provided the foxes initially are the cause of an instantaneous mortality rate of 1 or more in the mouse broods; and the greater the increase of the foxes, the less this initial mortality need be to accomplish the result stated. To put it another way, the effectiveness of foxes in reducing the survival of mice will increase more rapidly than does the actual number of foxes, whenever they are already reducing mouse survival to 37 per cent or less of what it would be in the absence of foxes. This effect gives the foxes an advantage which tends to counterbalance the greater rate of reproduction of the mice.

More generally, there is evidently no reason why a large predator, or assemblage of predators, might not regulate the abundance of a smaller and more prolific prey species, provided their abundance can change with that of the prey to *some* extent. Indeed, since a given change in predator abundance has a greater effect upon survival, the smaller the average survival rate of the prey (and hence

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the greater its fecundity), it could be argued that great fecundity should make a species more—not less—easily controlled by small changes in predator abundance. Against this must be laid the fact that great fecundity makes the prey more likely to occasionally "slip out from under" by becoming so numerous that the type B predation situation cannot be maintained.

Other Compensatory Agents

We have seen that both cannibalism and "ordinary" predation lead to the Figure 33 type of reproduction curve under type B conditions. For other compensatory agents only a qualitative examination can be attempted here, but they must be mentioned briefly.

The special type of predation situation in which insect "parasites" eat other insects regularly leads to low host reproduction at high host densities, as shown in Nicholson and Bailey's (1935) detailed analysis.

"True" parasites (those which do not regularly and consistently destroy their hosts) and disease organisms can also exhibit non-linear effectiveness which can seriously reduce reproduction at high stock densities—though their activity in this respect must usually be very irregular, if the human situation is a guide.

In the *Daphnia* cultures described earlier no predation or cannibalism existed, nor apparently did disease. The most plausible suggestion which has been made is that some metabolic product acted as a depressor of reproduction. If, for example, a metabolic waste killed a fixed fraction of *Daphnia* embryos per unit concentration and per unit time throughout prenatal life, it would have exactly the same effect upon survival as random cannibalism, and lead to the reproduction curve of Figure 33. The points of Figure 28 can be fitted by expression (10) with good apparent agreement.

Again, consider the compensatory situation where a brood, initially more numerous than average, grows more slowly than average because of competition for limited food; and because of this smaller individual size it is more vulnerable to predation than a faster growing brood at corresponding ages. It is fairly clear that here too there *could* be a lag effect whereby the brood would lose by increased predation more than the numerical advantage which it originally had; though it would be difficult to show that this should (or should not) occur as a general rule.

Finally, consider what happens when spawning facilities are limiting, as for some salmon or trout populations, for example. Natural selection will probably see to it that the population uses the best spawning facilities first. At increasing densities there is both a spread to less suitable gravel in other parts of the stream and a crowding and superimposition of eggs on the favourable grounds. With extremely heavy seeding, fungus may cover and kill almost the whole of the eggs in even the best redds; something of the sort probably decimated the Karluk pinks in the disastrous year 1924 (Barnaby, 1944). The sum of the above effects would usually add up to a domed reproduction curve with an extended right limb. Though an *exact* fit to expression (10) would not be likely, the latter might serve as a picture of the normal expectation.

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FITTING THE CURVE $z = we^{1-w}$

A first approximation to a fit of expression (10) to any body of data can be obtained very quickly, because the distribution is completely determined by the position of the maximum and the latter can be selected by eye. The work consists merely of dividing each observed estimate of reproduction (Z) by the estimated maximum Z, and each observed parental stock density W by the density which produced the estimated maximum W; this gives w and z, respectively. From (10),

$$\log_e \hat{z} = \log_e w + (1 - w). \tag{12}$$

A sample calculation is shown in the last line of Table VII.

To find the *best*-fitting curve of this type it would be necessary to calculate the residuals (differences between observed and calculated values) for a series of trial positions of the maximum point. The latter can be varied in two dimensions, so the process might be fairly protracted. The accepted criterion of best fit would be that for which the sum of the squares of the residuals is least, but an easier

TABLE VII. Computation of expected reproduction ($\hat{Z}$) and residuals (R) for a set of stock-reproduction (W–Z) observations, using $z = we^{1-w}$. W = 13, $Z_{max.} = 38$, is the trial position of the dome of the curve. The last line is a computation of $\hat{Z}$ for an arbitrary W. (See text for further details.)

1 W	2 Z	3 z	4 w	5 $\log_e z$	6 $\log_e w$	7 Unity	8 R	9 R^2	10 $\log_e \hat{z}$	11 $\hat{z}$	12 $\hat{Z}$
5	18	0.47	0.38	−0.75	−0.95	1.00	−0.42	0.18	−0.33	0.72	27
8	78	2.05	0.61	+0.72	−0.49	1.00	+0.82	0.67	−0.10	0.90	34
16	20	0.53	1.23	−0.64	+0.21	1.00	−0.62	0.38	−0.02	0.98	37
28	5	0.13	2.15	−2.02	+0.77	1.00	−1.64	2.69	−0.38	0.68	26
33	8	0.21	2.54	−1.56	+0.93	1.00	−0.95	0.90	−0.61	0.54	21
78	28	0.74	6.00	−0.30	+1.79	1.00	+2.91	8.47	−3.21	0.04	1.5
Sum	..	..	..	..	..	..	+0.10	13.29	..	..	..
50	..	..	3.85	..	+1.35	1.00	..	..	−1.50	0.22	8

criterion—that the sum of the residuals should approximate to zero—would be useful in the earlier stages. Table VII shows a computation of residuals (R) for the pink salmon data of Figure 19. The dome of the curve is estimated to be at a parental abundance of W = 13 and filial abundance of $Z_{max.} = 38$. Dividing the observed values of columns 1 and 2 by these values gives z and w in columns 3 and 4. The residual (R) is the observed value $\log_e z$ less the estimated $\log_e \hat{z}$ from (12), which is equivalent to

$$R = \log_e z + w - \log_e w - 1. \tag{13}$$

The four right-hand terms of (13) are in the order of columns 4–7 of Table VII; adding columns 4 and 5 and subtracting 6 and 7 gives R in column 8, and R^2 is shown in 9. The sum of the R's is close to zero, showing that the trial position of the dome of the curve cannot be too far from the best one; however since more than half of the R^2 total is contributed by the last observation, a somewhat better fit could probably be obtained by moving the dome a little to the right or upward.

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Expected reproduction values, $\hat{Z}$, are calculated in the last three columns of Table VII. Column 10 is equal to 8 less 5; or it can be obtained directly by adding columns 6 and 7 and subtracting 4 (cf. formula 12). The latter method is used to calculate $\hat{Z}$ for arbitrary values of W, as shown in the last line of the Table.

Base-10 logarithms can be used in Table VII if columns 4 and 7 are multiplied by 0.4343, but the procedure shown is more convenient when a comprehensive table of natural logarithms is at hand.

Fitted curves have not been used on the observed reproduction figures above (Figs. 18–28). Apart from the work involved, the theory of expression (10) needs further examination, and uncritical use of it now might conceal variant or alternative situations, such as that suggested for Figure 21. All the curves of Figures 18–28 were drawn freehand before expression (10) had been developed, and it is interesting that most of these observations suggested a concave and tapering right limb. Even the original hypothetical curves of Figures 7 and 8 were drawn this way, because of a feeling that reproduction couldn't really decline right to zero as spawners became increasingly numerous.

Returning to the question introduced at the start of this division of the paper, no universally applicable theory of reproduction has been discovered, or is likely to be. However several possible reasons are apparent why recruitment can decline, and usually will decline, at higher stock densities. Also, from reasonable assumptions a simple mathematical expression has been developed which could be used to represent most of the observed reproduction data over the range of densities which they cover.

SUMMARY

1. The general theory of reproduction indicates that density-dependent causes of mortality set a limit to the size which a population achieves. The "reproduction curve" for a fish species is defined as a graph of the average number of eggs produced by a filial generation against the number produced by its parental spawning assemblage, under the existing frequency distribution of environmental conditions for survival (Figs. 1–8).

2. The level of adult stock at which a *maximum number* of recruits (on the average for existing variations in environmental conditions) is obtained is not necessarily the level at which the *replacement number* is obtained (on the average). Maximum recruitment may occur either at the replacement level of adult stock (Figs. 2, 3), or at a higher level (Figs. 1, 4), or at a lower one (Figs. 5–8).

3. When, with increasing stock density, the maximum of recruits exceeds and precedes the replacement number, a population tends to oscillate in abundance. These oscillations are stable if the reproduction curve crosses the 45-degree line (representing the replacement level of reproduction) with a slope between -1 and $-\infty$ (Figs. 7, 8); they are damped if the curve crosses at any numerically lesser slope (Fig. 5).

4. Under the conditions of Figures 7 and 8 a population of single-age spawners has an irregular but permanent cycle of abundance, such as shown in Figure 11D for example, if enivronmental conditions are stable.

5. Under the same conditions a population of multiple-age spawners tends to have a more regular cycle, whose peak-to-peak period is close to twice the median length of time from oviposition by the parent generation to oviposition by all its progeny (Figs. 9F, 10).

6. For such populations to have cycles of appreciable amplitude, it is necessary that reproduction be absent or light during the first year or two of the animal's life (assuming that reproduction occurs once a year).

7. Random fluctuation in reproductive success, *by itself*, leads to very large changes in abundance and eventual extinction or fragmentation, for populations of single-age spawners (Fig. 11A), and to unlimited increase for multiple-age spawners (Fig. 12A).

8. Combinations of random fluctuation in reproduction with reproduction curves produce populations which neither increase indefinitely nor decline to extinction. The cyclical changes produced by steep reproduction curves are maintained when random fluctuation is small or moderate, but they become variable in period and eventually unrecognizable as random effects become more important.

9. Under the combined influence of a steep reproduction curve and variable non-compensatory mortality, populations fluctuate more widely than when controlled by either of these factors alone.

10. When the dome of a population's reproduction curve lies above the 45-degree line, the result of light or moderate exploitation is to *increase* the abundance of the stock in subsequent generations; more intensive exploitation will decrease it (Fig. 16).

11. Another result of exploitation is to reduce the amplitude and complexity of any reproduction-curve oscillations that may be in progress; sufficiently intensive exploitation eliminates such oscillation entirely (Fig. 17).

12. Among multiple-age spawners exploitation tends also to reduce the *period* of oscillation somewhat, because the spawning stock gradually becomes younger.

13. Reproduction curves, or approximations to them, are plotted for fish populations and four invertebrate populations in Figures 18–28. No example was discovered where the maximum of recruitment is at or to the right of the 45-degree line (as in Figs. 1, 3, 4). The left limb was usually steep. The most probable position for the right limb was always sloping downward, sometimes only slightly, sometimes quite steeply; the most typical situation has a slope about as in Figure 5.

14. Cyclic population changes that are apparently the direct result of a steep reproduction curve are those of Herrington's haddock (1912–29) and the *Daphnia* cultures of Pratt. Most other well-known cycles seem not to be of a *simple* reproduction-curve type, but the shape of this curve must profoundly influence the course of population abundance in any animal.

15. More complex reproduction curves have been suggested by indirect evidence (Figs. 30, 31), and a depression of percentage reproduction at extremely low densities of stock may be fairly common.

16. An asymmetrical dome-shaped reproduction curve can be developed from simple assumptions involving random cannibalism or compensatory preda-

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tion: if w represents adult stock density as a fraction or multiple of the density which provides maximum reproduction, we^{1-w} represents the actual reproduction at density w, as a fraction of the maximum reproduction. In this event the survival rate at maximum reproduction is always $1/e$ or 37 per cent of what it is at densities near zero; or in other words, at maximum reproduction the instantaneous mortality rate from compensatory activity is 1.

17. Curves computed from the above formula, shown in Figure 33, include most of the types actually observed. However if "environmental capacity" sets an upper limit to number of survivors at some stage of the pre-adult life history, this "typical" curve could either be levelled off at any point of its course, or else truncated (Fig. 34).

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